**GOVERNMENT OF GOA**

# FOURTH CENSUS OF MINOR IRRIGATION SCHEMES – GOA 2006-07



**DIRECTORATE OF PLANNING, STATISTICS AND EVALUATION**

**PANAJI - GOA**

# *PREFACE*

Agriculture is an important sector of the States economy contributing almost 6% of the Gross State Domestic Product at current prices for the year 2005-06. One of the essential input to increase the agricultural output is irrigation. Under the orbit of Irrigation, the Minor Irrigation Source plays an important role.

As the data available regarding the potential created by these Minor Irrigation sources by the various States was conflicting and efforts to reconcile the same were not successful, Government of India decided to conduct a Census of Minor Irrigation throughout the country with 100% Central Assistance on quinqunnial basis. Goa is also participating in this National level programme.

This report brings out the results of the 4th Minor Irrigation Census conducted as per the guidelines provided by the Ministry of Water Resources, Government of India. The report presents a brief analysis of the Census results both at the State and District level. The detailed source-wise information on different types of Minor Irrigation sources can be seen in **Annexure-I** as prescribed by the Ministry of Water Resources, Government of India.

This publication is the result of sincere and pains taking work done by a team of Enumerators, Supervisors, BDO Level Statistical Assistant – Coordinators, Additional Taluka Charge Officers, Taluka Charge Officers, the staff appointed in Minor Irrigation Cell at Head quarter and over all supervision by Joint Director.

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# CHAPTER – I
# BACKGROUND

1.1 Irrigation is essential for increasing the cropping intensity as also the land productivity. It also creates additional employment opportunities in the rural areas. The expansion and improvement of irrigation facilities is, therefore, a key ingredient of agricultural and rural development. The Major/Medium irrigation projects not only involve huge investments but also long gestation period. Further, they contribute to the displacement of large number of families, staying for generations in the catchments areas, from their ancestral homes/avocations and give rise to problems associated with their rehabilitation. Some projects in addition to providing water for irrigation also provide water for domestic and industrial purposes and quite often generate power. Minor Irrigation Projects on the other hand require relatively lesser investment with practically no gestation period and can be taken up with local resources without specialized technical skills. They also provide dispersed employment and yield quicker and wide spread results. Minor Irrigation, therefore, plays a crucial role in the development of irrigation potential of our country where the scarce resources of the state are under pressure from several directions.

## CENSUS OF MINOR IRRIGATION SCHEMES

1.2 Minor Irrigation Schemes include all schemes of Ground Water development and Surface Water development (flow and lift) which individually have a culturable command area of up to 2000 ha.

1.3 As part of various developmental programmes, several Minor Irrigation Schemes (MIS) have been implemented by various states. As there is no Act regulating the implementation of MIS, the same could have been implemented in the command area of major and medium irrigation projects resulting in the potential created by these MIS and their utilization being not accounted properly. A lot of aphorism has gone in the maintenance of statistics on this sector. Since, efforts to reconcile much of the conflicting data did not meet with success, Government of India decided upon conducting a complete enumeration of minor irrigation sources through

a census with 100% Central assistance to create a credible database for future development of this sector.

## MINOR IRRIGATION SOURCES IN GOA

1.4 At the time of Liberation in 1961, there were several sources of Minor Irrigation such as lakes, ponds, wells, fountains etc. There were also two small water reservoirs built by constructing dams at Khandepar across the river Khandepar and at Paroda across the river Kushavati, with Canal distribution systems, irrigating approximately 200 ha. and 380 ha. of land respectively. The total estimated area under irrigation from all these minor irrigation sources was 7860 ha. approximately.

1.5 With the implementation of various developmental schemes after liberation of the Territory, the total irrigation potential under Minor Irrigation has increased considerably. The 4th Census of Minor Irrigation conducted in year 2006-07 with reference year 2006-07 indicates that the irrigation potential by the end of 2006 was of the order of 12,606 ha. while the area actually irrigated was 11,853 ha. which is roughly 15% of the total cropped area during the year 2006-07. The Irrigation Department has tentatively assessed the potential for Minor Irrigation as 82,260 ha. under surface water sources and 7,400 ha. under ground water resources. There is, as such, ample scope for increasing the minor irrigation potential in Goa.

1.6 According to the Water Resource Department, Government of Goa, the irrigation potential created at the end of the Tenth Five Year Plan (2006-07) by way of Major and Medium Irrigation Projects was 19,977 ha. and by way of Minor Irrigation Projects was 28,827 ha. The total irrigation potential created was 52454 ha. (19977 + 28827) and the total utilization under Major & Medium Irrigation is 7700 ha. and Minor Irrigation is 27,426 ha. The contribution of Minor Irrigation in the total utilization was 35,126 ha. which amounts to 66.97% of the total utilization. Thus, Minor Irrigation Schemes still continue to be the predominant sources of irrigation in Goa.

1.7 In Chapter II of the report, a profile of Goa indicating the area, population, salient features of the agricultural economy and soil condition etc. is discussed. Chapter III contains the detailed concepts, scope and coverage of the Census. Chapter

IV contains the State and District level analysis of the Census results. The summary of important findings of the Minor Irrigation Census are presented in Chapter V. The Taluka wise data regarding the Census results can be found in the detailed tables appended as ANNEXURE I.

# CHAPTER – II

# GOA – A PROFILE

## 2.1 LOCATION AND CLIMATE

The Goa State lies between 14° and 16° North latitude and 73°and 75° East longitude. It is situated on the West Coast of India and is surrounded by the Arabian Sea on the west, Sindhudurg district of Maharashtra on the north, Sahyadri Mountains on the east and Uttara Kannada district of Karnataka on the south. For administrative purpose, the State is divided into two districts viz. North Goa District and South Goa District with 6 and 5 Talukas respectively. The climate is warm and humid. It receives heavy precipitation varying from 1500 to 3400 mm during the period from June to September from South-West monsoons. The maximum and minimum temperature ranges from 17.4ºC to 36ºC with relative humidity variation from 76% to 86%.

## 2.2 AREA AND POPULATION

The State covers an area of 3,702 sq. kms. with Sanguem Taluka in South Goa District occupying the highest coverage of 24 percent of the total area, followed by Sattari in North Goa, occupying 13 percent. The Mormugao Taluka in South Goa occupies the lowest area i.e. 3 percent.

## 2.3 DEMOGRAPHY

As per 2001 Population Census, the population of the State is 13,47,668 persons, with a density of 364 persons per sq. km. as against a density of 324 at all India level. Among the different Talukas of the State, the density of population is highest in Mormugao Taluka with 1328 persons per sq.km, followed by Salcete with 895 persons per sq.km. The Sanguem Taluka has the lowest density of 77 persons per sq.km. Out of the total population, 49.76% are found to be residing in urban areas. The female sex ratio is 961. The literacy percentage is quite high (82 %) with male literacy of 88.42 and female literacy of 75.49%. The salient features of area and population are as given below:-

**TABLE 2.1**

AREA AND POPULATION

| Sl. No. | Taluka | Area ($Km^2$) | Population | Density $Km^2$ | Percentage of urban population to total population |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| 1. | Tiswadi | 213.5 | 1,60,091 | 750 | 65.63 |
| 2. | Bardez | 264.0 | 2,27,695 | 863 | 58.61 |
| 3. | Pernem | 251.7 | 71,999 | 286 | 13.35 |
| 4. | Bicholim | 238.8 | 90,734 | 380 | 40.87 |
| 5. | Sattari | 495.1 | 58,613 | 120 | 13.51 |
| 6. | Ponda | 292.8 | 1,49,441 | 510 | 32.53 |
| 7. | Sanguem | 873.8 | 64,080 | 77 | 17.18 |
| 8. | Canacona | 352.0 | 43,997 | 125 | 27.05 |
| 9. | Quepem | 318.3 | 74,034 | 233 | 45.90 |
| 10. | Salcete | 292.9 | 2,62,035 | 895 | 57.85 |
| 11. | Mormugao | 109.1 | 1,44,949 | 1328 | 83.04 |
| | **Goa State** | **3702.0** | **13,47,668** | **364** | **49.76** |

Source: Population Census 2001

Note : Area figures for the Districts & State were supplied by the Surveyor General of India and those for the Talukas & Towns were supplied by the Directorate of Settlement of Land Records. The total Taluka areas will not tally with the respective District totals.

## 2.4 LAND UTILIZATION

The following table gives the details of land utilization in the State.

**TABLE 2.2**

LAND UTILIZATION

| Sl. No. | Land Use Category | Area in hectares | Percentage to total geographical area |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
| 1. | Total geographical area according to Land Survey Department | 3,61,113 | 100.00 |
| 2. | Area under Forest | 1,25,473 | 34.75 |
| 3. | Land not available for cultivation | 37,137 | 10.28 |
| 4. | Other uncultivable land | 61,722 | 17.09 |
| | i) Permanent Pastures & other grazing lands | 1,305 | 0.36 |
| | ii) Land under miscellaneous tree crops and groves not included in net area sown | 580 | 0.16 |
| | iii) Cultivable waste (including fallow land) | 59,837 | 16.57 |
| 5. | Net area sown | 1,36,781 | 37.88 |
| 6. | Area sown more than once | 33,546 | - |
| 7. | Gross cropped area | 1,70,203 | - |

Source: Statistical Pocket Book, 2005-06

Though the total geographical area according to the survey of India is 3,70,200 ha. the same according to the Land Survey Department is 3,61,113 ha. The area under forest is 34.75% while net sown area is 37.88%. A vast area of land, comprising of 16.57% is cultivable waste, which includes fallow land also. About 10.28% of land is not available for cultivation. The area sown more than once is nearly 9.29%.

## 2.5 LAND HOLDINGS

The distribution of number of holdings by size classes along with its percentages as recorded in the Agricultural Census 2005-06, are given below:

**TABLE 2.3**

NUMBER AND AREA OF OPERATIONAL HOLDINGS

| Sl. No. | Size classes (ha.) | Total holding | |
|---|---|---|---|
| | | Number | Area (ha.) |
| 1 | 2 | 3 | 4 |
| 1 | Below 0.5 | 33184 (62.82) | 6226 (10.25) |
| 2 | 0.5 – 1.0 | 9561 (18.10) | 6196 (10.20) |
| 3 | 1.0 – 2.0 | 5788 (10.96) | 7184 (11.83) |
| 4 | 2.0 – 3.0 | 1990 (3.77) | 4470 (7.36) |
| 5 | 3.0 – 4.0 | 691 (1.31) | 2255 (3.71) |
| 6 | 4.0 – 5.0 | 449 (0.85) | 1902 (3.13) |
| 7 | 5.0 – 7.5 | 524 (0.99) | 3046 (5.01) |
| 8 | 7.5 – 10.0 | 222 (0.42) | 1865 (3.07) |
| 9 | 10.0 – 20.0 | 282 (0.53) | 3667 (6.04) |
| 10 | 20.0 and above | 130 (0.25) | 23931 (39.40) |
| | **Total** | **52821 (100)** | **60742 (100)** |

Source: Agricultural Census 2005-06
Note: Figures in brackets indicate percentages

There are 52821 operational landholders with total holding area of 60742 ha.. The average size of the holding is 1.15 ha. The proportion of landholders below 0.5 ha. is

63 percent, between 0.5 and 1.0 ha. is 18% percent and between 1.0 ha. and 2.0 ha. is 11%. Thus it can be seen that about 92% land holders are holding land of less than 2.0 ha., the average area of the holding being 0.40 ha. Whereas the remaining 8 percent land holders have holding of an average size of 9.59 ha. It is thus evident that the land distribution in Goa is very uneven.

## 2.6 SOIL

The region has undulating hilly terrain with diverse soil conditions especially on its eastern side, where lies the southern end of the Sahyadri ranges. The terrain is intersected by a number of rivers flowing westwards, which provides a network of internal waterways. Three types of soils viz. i) Laterite or lateritic, ii) Alluvial and iii) Sandy are found in the region. About 2,75,900 ha. (81 per cent of the total area) are of the first category and are highly acidic in nature, sandy loam to silt loam in texture and well drained. They are poor in lime, Phosphorus and Potash, but fairly good in organic matter and Nitrogen. About 24,950 ha. (7.4 per cent of the total area) is alluvials, 17,600 ha. of which are subject to inundation of saline water and have to be protected by bunds to grow crops successfully. The nine kilometers coastal inland comprises of a stretch of 38,000 ha. (11.2 percent of the total area) half of which has a high water table and could be exploited for irrigation. These soils are also acidic, sandy to sandy loams, fairly rich in organic matter but very deficient in Phosphate and Potash.

## 2.7 CROPPING PATTERN

The State has been blessed with such soils and climatic conditions that it favours growth of a variety of tropical crops. Rice is the principal crop, which covers 52,177 ha. accounting for about 30.66% of the total cropped area. It is grown both in Kharif and Rabi seasons. Ragi is another cereal which occupies nearly 441 ha. or about 0.27 %. The area under Ragi is found to be declining over the years. Area under pulses is 10,978 ha. or 6.45 percent which are grown mainly in Rabi season – on residual moisture after Kharif paddy. Sugarcane covers 1,110 ha. which is also about 1%. Cashewnut and coconut are another two important cash crops which occupy nearly 55,021 ha. and 25,312 ha. or 32 and 15 % respectively. The details of cropping pattern are given below:

**TABLE – 2.4**

ESTIMATES OF TOTAL AREA UNDER PRINCIPAL CROPS

| Sl. No. | Crops | Total area under the crop (ha.) | Percentage to the total cropped area |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
| 1 | Rice | 52177 | 30.66 |
| | i)Kharif | 35112 | 20.63 |
| | ii)Rabi | 17065 | 10.03 |
| 2 | Ragi | 441 | 0.26 |
| 3 | Maize | 100 | 0.06 |
| 4 | Pulses | 10978 | 6.45 |
| 5 | Groundnut | 3312 | 1.94 |
| 6 | Arecanut | 1641 | 0.96 |
| 7 | Coconut | 25312 | 14.87 |
| 8 | Cashewnut | 55021 | 32.33 |
| 9 | Sugarcane | 1110 | 0.65 |
| 10 | Mango & Banana | 6564 | 3.86 |
| 11 | Pineapple | 318 | 0.19 |
| 12 | Vegetables | 8144 | 4.78 |
| 13 | Other fruits | 3645 | 2.14 |
| 14 | Oil palm | 814 | 0.48 |
| 15 | Pepper | 626 | 0.37 |
| | **Area under all crops** | **1,70,203** | **100.00** |

Note : Estimated figures relates to 2005-06

Source: Statistical Handbook of Goa 2005-06.

## 2.8 CONTRIBUTION OF AGRICULTURE TO STATE INCOME

Though agriculture proper is not the predominant sector in the State it contributes a sizeable share to the State income estimate. The share of Agriculture in the Gross State Domestic Product of Goa State for the year 2006-07 works out to Rs.762.44 crores or 4.61% at the current trends. At constant (2004-05) prices, this share works out to Rs.650.81 crores, which record 4.33% of GSDP at Constant Prices.

# CHAPTER – III

## CONCEPTS, SCOPE & COVERAGE

### 3.1 SCOPE AND COVERAGE

The Census of Minor Irrigation 2006-07 was conducted during the period December 2007 to May 2008. The reference period was the agricultural year 2006-07. The Census was conducted throughout the country for the 4th time. All the 388 revenue villages of this State were covered under the Minor Irrigation Census as also 194 urban wards for all the urban areas. The data was collected in respect of each of the sources of Minor Irrigation located in the villages and in urban areas. Minor Irrigation sources include Dug wells, Shallow Tube wells and Deep Tube wells meant for irrigation. Minor Irrigation sources also included Surface Lift Irrigation Schemes and Surface Flow Irrigation Schemes.

### 3.2 DEFINITION OF MINOR IRRIGATION SCHEMES

Irrigation Schemes having a Culturable command area upto 2000 ha. individually are classified as Minor Irrigation Schemes.

### 3.3 OBJECTIVES OF THE CENSUS

1. To enumerate all the sources of Minor Irrigation in the State of Goa.
2. To assess source-wise area irrigated during Kharif, Rabi and Summer seasons.
3. To assess the contribution of these Minor Irrigation sources by way of new irrigation and as supplementary irrigation sources.

### 3.4 ORGANIZATION OF WORK

Primary enumeration work was entrusted to the field workers from the Directorate of Planning, Statistics & Evaluation, Directorate of Agriculture, Water Resources Department and Block Development Offices. The supervision of the field work was given to the Research Assistants and Statistical Officers.

The enumerators were required to prepare a complete list of all the sources of Minor Irrigation in the villages by visiting and contacting knowledgeable persons like Sarpanchas, Ward Members, Gram Sevaks, Village Panchayat Secretaries, etc. Detailed enquiry was made to locate each source of Minor Irrigation. Besides, as per

the guidelines laid down by the Ministry of Water Resources, Government of India, in addition to enumeration, sample checks were conducted under the supervision of the Directorate as Nodal Department by covering 10 per cent of the villages for detecting under enumeration or over enumeration and obtain correction factor for the Main Census results. The Officers of this Directorate were appointed to carry out these sample checks by visiting 39 villages selected by Systematic Sampling Method. To locate all the sources of Minor Irrigation, the lists were prepared through verbal enquiry or by house to house visit. However, enumerators and supervisors made special all out efforts to locate the sources. The owners of the Minor Irrigation sources were farmers/Cooperative Societies/Government Departments/Private Organizations/Groups of farmers, etc.

Being a Nodal Department, this Directorate conducted the training cum-workshop at the State headquarters where the Officers of this Directorate who were appointed as Additional Taluka Charge Officers in various Talukas were trained. At the Taluka level, training was given to the primary enumerators and supervisors by its Additional Charge Officers appointed by the Nodal Department. All the 11 Talukas were covered by appointing Additional Charge Officer and training was imparted to explain the schedules and procedures for filling the schedules. The supervisors cross checked the data of the enumerators while the field work was in progress. The Enumerators while filling up the schedules also interviewed the owners of the Minor Irrigation Schemes and collected the information on the basis of the enquiry made from them.

The physical verification of the schemes was also done by the enumerators. The purpose of the Census was explained to the farmers and assurance was given to them that the data furnished by them will be kept confidential. To conduct the survey, 100% central assistance was provided by the Government of India. The Block Development Officers compiled the information in the prescribed formats-Schedule 1,2,3 and furnished to this Directorate which consolidated the data for North Goa and South Goa Districts and for the State of Goa and passed it on to the Minor Irrigation Census Commissioner for releasing reports at the All India level.

## 3.5 INSTRUCTIONS FOR FILLING UP OF THE SCHEDULES

The schedules and instructions for conducting the Minor Irrigation Census were drafted by the Directorate of Planning, Statistics & Evaluation on the basis of model schedule and other material provided by the Ministry of Water Resources, Minor Irrigation Division, Government of India. These schedules to be canvassed by the enumerators were discussed thoroughly during training. In order to obtain complete and realistic data, a press note was released through the Directorate of Information in the local dailies giving general information about the conduct of the Minor Irrigation Census, 2006-07 in the State, personnel engaged, type of enquiry proposed, etc. and requesting co-operation from the public to give the required information correctly.

There were in all three types of schedules designed for collecting the required data under this Census. They were as follows:-

| | |
|---|---|
| Schedule - 1 | Ground water schemes |
| Schedule - 2 | Surface water schemes |
| Schedule -3 | Village Schedules |

Copies of the schedules are placed at Annexure-II

# CHAPTER – IV

# ANALYSIS OF THE CENSUS RESULTS

The Census covered all Minor Irrigation Schemes in the State. The data collected during the Census operation for the year 2006-07 have been tabulated as per the tabulation plan provided by the Ministry of Water Resources, Minor Irrigation Division, Govt. of India. In all 56 main tables have been prepared as per the design and appended to the report.

4.2 The sources of Minor Irrigation Schemes identified in the state during the year 2006-07 have been shown District-wise in the table below:

**TABLE 4.1**

NUMBER OF MINOR IRRIGATION SCHEMES (DISTRICTWISE ) IN 2006-07 BY TYPE

| Sl. No. | Type of Source | Commissioned upto 2001 (3rd Census) | | | Total number of schemes (Enumerated during 4th Census) | | |
|---|---|---|---|---|---|---|---|
| | | N. Goa | S. Goa | Total | N. Goa | S. Goa | Total |
| 1 | Dug wells | 2412 | 934 | 3346 | 2935 (68.7) | 1336 (31.3) | 4271 (100) |
| 2 | Shallow Tube wells | 26 | 42 | 68 | 35 (33.3) | 70 (66.7) | 105 (100) |
| 3 | Deep Tube wells | 14 | 17 | 31 | 25 (53.2) | 22 (46.8) | 47 (100) |
| 4 | Surface Flow Irrigation | 763 | 972 | 1555 | 848 (48.7) | 893 (51.3) | 1741 (100) |
| 5 | Surface Lift Irrigation | 260 | 501 | 761 | 346 (38.0) | 564 (62.0) | 910 (100) |
| | **All Sources** | **3475** | **2286** | **5761** | **4189 (59.2)** | **2885 (40.8)** | **7074 (100)** |

Note:- Figures in brackets indicate percentage.

4.3 In all 7074 schemes of Minor Irrigation were enumerated during the course of field operation, out of which 5761 (81%) were found to have been commissioned upto the year 2000-01. The remaining 1313 schemes (19%) were commissioned during the period from 2002 to 2007.

4.4 Further, analysis of the above table shows that in terms of number of Minor Irrigation Schemes, Dug wells emerge as the most common source of Irrigation with 4271 wells in the State including 3346 wells or 78% commissioned upto 2000-01. The maximum number of wells i.e. 2935 (68%) were located in North Goa District, out of which 2412(82%) wells were commissioned upto 2001. In South Goa District,

on the other hand there were 1336 wells out of which 934 (70%) were commissioned upto 2000-01.

4.5 The surface flow irrigation forms the next common and important source with 1741 schemes including 1555 (89%) schemes commissioned upto 2000-01. The district of South Goa had the maximum number of schemes i.e. 893 (51%) as against 848 (49%) in North Goa District. The majority of schemes in both the districts were commissioned upto 2000-01.

4.6 The lift Irrigation is also found to be a significant source with 910 schemes out of which 761(84%) schemes were commissioned upto 2000-01. The majority of schemes are located in South Goa i.e. 564 (62%) schemes against 346 (38%) schemes in North Goa District.

4.7 In all 105 shallow tube wells were covered under the Census with 35 tube wells located in North Goa and 70 tube wells in South Goa District. While 42 tube wells in South Goa and 26 wells in North Goa were commissioned upto 2000-01. 47 deep tube wells were constructed in the State, of which 25 deep tube wells were found in North Goa and 22 deep tube wells in South Goa. Out of 25 Deep Tube wells 14 in North Goa District and only 17 wells out of 22 in South Goa District were commissioned upto 2001.

4.8 It has been observed that 463 Minor Irrigation Schemes over a period, have been recorded under 'Not in Use' category due to various reasons. It was, therefore, a significant aspect of the Census to locate and find out such schemes under use and not in use. The data collected in the respect has been presented in the table 4.2.

**TABLE 4.2**

NUMBER OF WELLS BY TYPE INUSE AND NOT INUSE

| Sl. No. | Type of Source | In Use | | | Not in Use | | | Total | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | North Goa | South Goa | Goa State | North Goa | South Goa | Goa State | North Goa | South Goa | Goa State |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
| 1 | Dug wells | 2691 (63.0) | 1263 (29.6) | 3954 (92.6) | 244 (5.71) | 73 (1.7) | 317 (7.4) | 2935 (68.7) | 1336 (31.3) | 4271 (100.0) |
| 2 | Shallow Tubewells | 30 (28.6) | 54 (51.4) | 84 (80.0) | 5 (4.8) | 16 (15.2) | 21 (20.0) | 35 (33.3) | 70 (66.7) | 105 (100.0) |
| 3 | Deep Tubewells | 25 (53.2) | 13 (27.7) | 38 (80.9) | 0 (0.0) | 9 (19.1) | 9 (19.1) | 25 (53.2) | 22 (46.8) | 47 (100.0) |
| 4 | Surface flow Irrigation | 782 (44.9) | 861 (49.5) | 1643 (94.4) | 66 (3.8) | 32 (1.8) | 98 (5.6) | 848 (48.7) | 893 (51.3) | 1741 (100.0) |
| 5 | Lift Irrigation | 338 (37.1) | 554 (60.9) | 892 (98.0) | 8 (0.9) | 10 (1.1) | 18 (2.0) | 346 (38.0) | 564 (62.0) | 910 (100.0) |
| | **All Sources** | **3866 (54.7)** | **2745 (38.8)** | **6611 (93.5)** | **323 (4.5)** | **140 (2.0)** | **463 (6.5)** | **4189 (59.2)** | **2885 (40.8)** | **7074 (100.0)** |

4.9 It can be seen from the above table that 93.5% of the Minor Irrigation Schemes were in use. Only 6.5% of the schemes were not in use in the State of which 4.5% and 2% in North and South Districts respectively.

4.10 Source wise analysis reveals that out of 105 shallow tube wells, 05 shallow tube wells were not in use in North Goa District and 16 in South Goa District and out of 47 Deep tube wells 9 were not in use in South Goa District. Against 1741 surface flow and 910 surface lift irrigation schemes, 1643(94.4%) and 892 (98%) respectively were in use. Further 782 (44.9%) surface flow schemes and 338 (37.1%) surface lift irrigation schemes were in use in North Goa District as compared to 861 (49.5%) schemes and 554 (60.9%) schemes respectively in South Goa District. It may also be seen from the table above, that out of 4271 Dug wells in the State 3954 (92.6) dug wells were in use, being 2691 (63.0%) dug wells in North Goa District and 1263 (29.6%) dug wells in South Goa District.

4.11 The size of the holding gives an idea of average unit of cultivation ha. in the State of Goa. In particular where large portion of cultivated area is rain fed and scope of using

sub-soil water for irrigation is rather limited. The presence of any source of irrigation within or nearby to the operational holding enhances its productivity with multi-cropping system. The table No.4.3 below presents the distribution of all Minor irrigation sources according to the size of operational holdings in respect of State and Districts.

**TABLE 4.3**

DISTRIBUTION OF DUG WELLS AND SHALLOW TUBE WELLS ACCORDING TO SIZE OF OPERATIONAL HOLDINGS BY INDIVIDUAL FARMERS

| Sl. No. | Source | State/District | Size of holding (ha.) | | | | | Total (4 to 9) |
|---|---|---|---|---|---|---|---|---|
| | | | Marginal 0 – 1 | Small 1 - 2 | Semi Medium 2-4 | Medium 4-10 | Large (Big) >=10 | |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| 1 | Dug wells | Goa State | 3131 (73.3) | 549 (12.9) | 406 (9.5) | 119 (2.8) | 66 (1.5) | 4271 (100.0) |
| | | North Goa | 2370 (55.5) | 283 (6.6) | 163 (3.8) | 85 (2.0) | 34 (0.8) | 2935 (68.7) |
| | | South Goa | 761 (17.8) | 266 (6.2) | 243 (5.7) | 34 (0.8) | 32 (0.8) | 1336 (31.3) |
| 2 | Shallow tube wells | Goa State | 59 (56.2) | 16 (15.2) | 20 (19.0) | 8 (7.7) | 2 (1.9) | 105 (100.0) |
| | | North Goa | 20 (19.0) | 5 (4.8) | 6 (5.7) | 3 (2.8) | 1 (1.0) | 35 (33.3) |
| | | South Goa | 39 (37.1) | 11 (10.5) | 14 (13.3) | 5 (4.8) | 1 (1.0) | 70 (66.7) |
| 3 | Deep Tube Wells | Goa State | 25 (53.2) | 7 (14.9) | 4 (8.5) | 3 (6.4) | 8 (17.0) | 47 (100.0) |
| | | North Goa | 6 (12.8) | 5 (10.6) | 3 (6.4) | 3 (6.4) | 8 (17.0) | 25 (53.2) |
| | | South Goa | 19 (40.4) | 2 (4.3) | 1 (2.1) | 0 (0.0) | 0 (0.0) | 22 (46.8) |
| 4 | Surface Flow Schemes | Goa State | 1414 (81.2) | 146 (8.4) | 142 (8.2) | 30 (1.7) | 9 (0.5) | 1741 (100.0) |
| | | North Goa | 757 (43.5) | 47 (2.7) | 29 (1.7) | 11 (0.6) | 4 (0.2) | 848 (48.7) |
| | | South Goa | 657 (37.7) | 99 (5.7) | 113 (6.5) | 19 (1.1) | 5 (0.3) | 893 (51.3) |
| 5 | Surface Lift Schemes | Goa State | 314 (34.5) | 187 (20.6) | 237 (26.0) | 142 (15.6) | 30 (3.3) | 910 (100.0) |
| | | North Goa | 229 (25.2) | 36 (4.0) | 35 (3.8) | 40 (4.4) | 6 (0.7) | 346 (38.1) |
| | | South Goa | 85 (9.3) | 151 (16.6) | 202 (22.2) | 102 (11.2) | 24 (2.6) | 564 (61.9) |

Note: Figures in brackets indicate percentages.

4.12 It may be seen from the table above that 3131 (73.3%) dug wells were located in the operational holdings of the size less than 1 ha. as compared to 549 (12.9%) in the size of holding between 1 to 2 ha. and 591 (13.8%) dug wells in the size class 2 ha. and above. Majority of the wells i.e. 2370 (55.5%) in the size class below 1 ha. were found in North Goa as compared to 761 (17.8%) wells in South Goa District. Number of wells in the size class of operational holdings above 1 ha. accounted for 565 (13.2%) in North Goa District as against 575 (13.5%) in South Goa District.

4.13 Out of 105 shallow tube wells in the State, 59(56.2%) were in the size class below 1 ha., 16 (15.2%) were in size class of 1 to 2 ha. and 30 (28.6%) were in the size class of 2 ha. and above. While 20 (19.0%), 5 (4.8%) and 10 (9.5%) shallow tube wells in the size class of below 0-1, 1-2 ha, & 2& above and 39 (37.1%), 11(10.5% and 20(19.1%) tube wells of the above size class were in South Goa District. During the course of field work data regarding ownership of ground water sources according to social groups was also collected. The same has been tabulated and presented in the following table.

4.14 It may be seen from the above table that 25(53.2%) Deep Tube well were located in the operational holding of the size less than 1 Ha. as compared to 7 (14.9%) in the size holding between 1 to 2 ha and a total 15(31.9%) Deep tube wells in the size class 2 Ha. and above. Majority of the wells i.e 19(40.4%) in the size class below 1 Ha. were found in South Goa as compared 6(12.8%) wells in North Goa District. No of wells in the size class of operations holding above 1 Ha. accounted for 19(46.8%) in North Goa District as against 3(6.3% ) in South Goa District.

4.15 It may be seen from the above table that 1414(81.2%) Surface Flow Scheme were located in the operational holding of the size less than 1 Ha. as compared to 146 (8.4%) in the size holding between 1 to 2 ha and a total 181(10.4%) Surface Flow Scheme in the size class 2 Ha. and above. Majority of the Scheme i.e 757(43.5%) in the size class below 1 Ha. were found in North Goa as compared 657(37.7%) Scheme in South Goa District. No of Scheme in the size class of operations holding above 1 Ha. accounted for 91(5.2%) in North Goa District as against 236(13.6%) in South Goa District.

4.16 It may be also seen from the above table that 314(34.5%) Surface Lift Scheme were located in the operational holding of the size less than 1 Ha. as compared to 187 (20.6%) in the size holding between 1 to 2 ha and a total 409(44.9%) Surface Lift Scheme in the size class 2 Ha. and above. Majority of the Scheme i.e 229(25.2%) in the size class below 1 Ha. were found in North Goa as compared 85(9.3%) Scheme in South Goa District. No of Scheme in the size class of operations holding above 1 Ha. accounted for 117(12.9%) in North Goa District as against 479(52.6%) in South Goa District.

4.17 It may be seen from the below table No. 4.4 that only 100 (2.4%) dug wells were owned by Scheduled Castes. It may be pertinent to note here that the SC population of Goa is hardly 2%.

**TABLE 4.4** DISTRIBUTION OF GROUND WATER SOURCES OWNERSHIP ACCORDING TO SOCIAL STATUS

| Sl. No. | Source | State/District | Ownership according to Social status | | | | Other than Individual | Total (4 to 7) |
|---|---|---|---|---|---|---|---|---|
| | | | SC | ST | OBC | Others | | |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| 1 | Dug wells | Goa State | 100 (2.4) | 741 (17.4) | 698 (16.3) | 2282 (53.4) | 450 (10.5) | 4271 (100) |
| | | North Goa | 37 (0.9) | 337 (7.9) | 612 (14.3) | 1653 (38.7) | 296 (6.9) | 2935 (68.7) |
| | | South Goa | 63 (1.5) | 404 (9.5) | 86 (2.0) | 629 (14.7) | 154 (3.6) | 1336 (31.3) |
| 2 | Shallow tube wells | Goa State | 0 (0.0) | 28 (26.7) | 5 (4.8) | 31 (29.5) | 41 (39.0) | 105 (100) |
| | | North Goa | 0 (0.0) | 3 (2.9) | 3 (2.9) | 21 (20.0) | 8 (7.6) | 35 (33.3) |
| | | South Goa | 0 (0.0) | 25 (23.8) | 2 (1.9) | 10 (9.5) | 33 (31.4) | 70 (66.7) |
| 3 | Deep tube wells | Goa State | 0 (0.0) | 4 (8.5) | 0 (0.0) | 24 (51.1) | 19 (40.4) | 47 (100) |
| | | North Goa | 0 (0.0) | 2 (4.3) | 0 (0.0) | 19 (40.4) | 4 (8.5) | 25 (53.2) |
| | | South Goa | 0 (0.0) | 2 (4.3) | 0 (0.0) | 5 (10.6) | 15 (31.9) | 22 (46.8) |
| 4 | Surface Flow Schemes | Goa State | 49 (2.8) | 351 (20.2) | 87 (5.0) | 322 (18.5) | 932 (53.5) | 1741 (100) |
| | | North Goa | 12 (0.7) | 99 (5.7) | 67 (3.8) | 149 (8.6) | 521 (29.9) | 848 (48.7) |
| | | South Goa | 37 (2.1) | 252 (14.5) | 20 (1.2) | 173 (9.9) | 411 (23.6) | 893 (51.3) |
| 5 | Surface Lift Schemes | Goa State | 8 (0.9) | 220 (24.2) | 24 (2.6) | 538 (59.1) | 120 (13.2) | 910 (100) |
| | | North Goa | 3 (0.3) | 18 (2.0) | 19 (2.1) | 214 (23.5) | 92 (10.1) | 346 (38.0) |
| | | South Goa | 5 (0.5) | 202 (22.2) | 05 (0.6) | 324 (35.6) | 28 (3.1) | 564 (62.0) |

4.18 The data regarding Culturable Command Area, Gross Irrigation Potential created and Area irrigated by the various Minor Irrigation sources have been presented in the table below:

**TABLE 4.5**

DISTRIBUTION OF CULTURABLE COMMAND AREA, GROSS IRRIGATION POTENTIAL VREATED AND AREA IRRIGATED ACCORDING TO MINOR IRRIGATION SOURCES

| Sl. No. | Sources | Culturable Command Area (Ha.) | | | Gross Irrigation Potential Created (Ha.) | | | Area Irrigated (Ha.) | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | North Goa | South Goa | Goa State | North Goa | South Goa | Goa State | North Goa | South Goa | Goa State |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
| 1 | Dug wells | 2356 (48.4) | 2510 (51.6) | 4866 (100) | 2417 (48.3) | 2590 (51.7) | 5007 (100) | 2176 (46.7) | 2489 (53.3) | 4665 (100) |
| 2 | Shallow tube wells | 51 (38.1) | 83 (61.9) | 134 (100) | 53 (32.3) | 111 (67.7) | 164 (100) | 43 (39.4) | 65 (60.6) | 108 (100) |
| 3 | Deep tube wells | 173 (87.3) | 24 (12.2) | 197 (100) | 173 (85.2) | 30 (14.8) | 203 (100) | 173 (93.0) | 13 (7) | 186 (100) |
| 4 | Surface Flow Irrigation | 2199 (56.9) | 1669 (43.1) | 3868 (100) | 2270 (56.2) | 1771 (43.8) | 4041 (100) | 2112 (55.9) | 1669 (44.1) | 3781 (100) |
| 5 | Surface Lift Irrigation | 1265 (40.1) | 1868 (59.6) | 3133 (100) | 1315 (41.2) | 1877 (58.8) | 3192 (100) | 1278 (41.0) | 1835 (59.0) | 3113 (100) |
| | **All Sources** | **6044 (49.5)** | **6154 (50.5)** | **12198 (100)** | **6228 (49.4)** | **6379 (50.6)** | **12607 (100)** | **5782 (48.8)** | **6071 (51.2)** | **11853 (100)** |

4.19 **Culturable Command Area:** The above table No. 4.5 shows that Dug well irrigation formed an important and major source of minor irrigation with Culturable Command Area (CCA) of 4866 ha. (39.9%) as against the total area of 12198 ha. for the State, followed by Surface Flow irrigation with 3868 ha. (31.7%) and Surface lift irrigation with 3133 ha. (25.7%). The share of shallow and deep tube wells was negligible being mere 134 ha. and 197 ha. respectively.

4.20 The District wise analysis indicates that South Goa district led North Goa District with 6154 ha. (50.5%) of Culturable command area as compared to 6044 ha. (49.5%) in North Goa District. It can also be seen from the above table that Dug wells formed the predominant source in both the districts with 2356 ha. (19.3%) in North Goa District and 2510 ha. (20.6%) in South Goa District. The Surface Flow with 2199 ha. (18.0%) constitutes the next important source in North Goa District, while lift irrigation with 1867 ha. (15.3%) of CCA formed the second important source in South Goa District. The Surface Lift schemes in North Goa and Surface Flow Irrigation in South Goa district

ranked third with 1265 ha. (10.4%) and 1669 ha. (13.7%) of CCA's respectively. There is small CCA under Deep Tube well and shallow tube wells in North districts being 173 ha. (1.4%) and 51 ha.(0.4) While shallow tube well 83 h.a (0.7 %) and 24 h.a ( 0.2%) in South Goa District.

4.21 **Gross Irrigation Potential:** Further review of the above table indicates that as against 12607 ha. of irrigation potential created in the State, the utilization or area irrigated formed 11853 ha. (94%). While utilization rate in both the districts emerges with 5782 ha. (92.8%) area irrigated as against 6228 ha. potential created, in North Goa District and 6071 ha. (95.2%) area irrigated as against 6379 ha. irrigation potential created in South Goa. The source-wise analysis of Dug wells shows better utilization rate with 4665 ha. (93.2%) area irrigated as against 5007 ha. potential created followed by Surface flow irrigation with 3781 ha. (93.6%) against 4041 ha. potential created. Surface Lift Irrigation ranks third in the State with (97.5%) utilization or 3113 ha. irrigated against 3192 ha. potential created. Utilization under shallow & deep tube wells is negligible.

4.22 In South Goa District dug wells show higher percentage utilization as compared to wells in North Goa District. In South Goa 2489 ha. (96.1%) area is irrigated as against 2590 ha. potential created as compared to 2176 ha. (90.0%) area irrigated against 2417 potential created in North Goa District.

4.23 **Area Irrigated:** The above table shows that, in terms of area irrigated the Dug Wells Schemes constitute the most important source of irrigation followed by Surface Flow Irrigation Schemes and Surface Lift Schemes Dug wells. In the two districts also, the Dug Well Schemes constitute the most important source of irrigation. The second and third position in North Goa district is taken by Surface Flow and Surface Lift Irrigation schemes respectively whereas in South Goa district the second position is taken by the Surface Lift Irrigation Schemes and the third by Surface Flow Schemes. The area irrigated by Shallow and Deep Tube Wells is negligible. The total area irrigated is 11853 ha. (94%) of the irrigation potential created. Steps may, therefore, be taken for fuller utilization of the irrigation potential created through the Minor Irrigation Schemes.

# CHAPTER V

## COMPARATIVE STUDY OF CURRENT CENSUS WITH THE PREVIOUS CENSUS

5.1 **Number of Sources***:* In the below table, the total number of Minor Irrigation Schemes recorded in the State during the 4th Census were found to be 7074 as compared to 10051 of the previous census. Thus there has been a decrease of 29.6% in the number of the total sources. The decline in the number of Minor Irrigation Schemes observed in the current Census can be attributed to transfer of Agricultural land to Non-Agricultural uses including construction, the costs on factors of production in Agriculture continues to increase so also traditional factors of production such as Bullocks, Plough, etc., are reducing in quantity therefore the small and marginal farmers find it difficult to plough their fields resulting in non-commissioning of irrigation schemes. Surface irrigation sources drying up due to frequent cutting of surrounding trees as less trees bring less rain, so also young generation preferring jobs other than agricultural activities for which increased level of education also plays a part, are some of the reasons. The same trend was observed in both the districts. The total number of minor irrigation sources recorded in North and South Goa districts during the fourth Census were 4189 and 2885 respectively as against the 3rd Census figures of 7057 and 2994. There has been thus 40.64% and 3.64% decrease in the number of sources respectively in North and South as compared to the previous census conducted in 2007.

**TABLE – 5.1**

COMPARATIVE STATEMENT OF NUMBER OF IRRIGATION SOURCES IN THE FIRST, SECOND, THIRD AND FOURTH CENSUS

| State/District | Census | Ground Water Sources | Surface Irrigation | Total |
|---|---|---|---|---|
| **Goa State** | 1st Census | 5212 | 3764 | 8976 |
| | 2nd Census | 4022 | 3928 | 7950 |
| | 3rd Census | 5206 | 4845 | 10051 |
| | 4th Census | 4423 | 2651 | 7074 |
| | *****Variation (%) | (-15) | (-45.3) | (-29.6) |
| **North Goa District** | 1st Census | 4133 | 1786 | 5919 |
| | 2nd Census | 2891 | 2744 | 5635 |
| | 3rd Census | 3819 | 3238 | 7057 |
| | 4th Census | 2995 | 1194 | 4189 |
| | *****Variation (%) | (-21.6) | (-63.1) | (-40.6) |
| **South Goa District** | 1st Census | 1079 | 1978 | 3057 |
| | 2nd Census | 1131 | 1184 | 2315 |
| | 3rd Census | 1387 | 1607 | 2994 |
| | 4th Census | 1428 | 1457 | 2885 |
| | *****Variation (%) | (+3.0) | (-9.3) | (-3.6) |

*****Variation is as compared to 3rd Minor Irrigation Census.

A comparative statement of taluka wise irrigation sources is presented in the Annexure–A

5.2 **Ground Water Sources:** It has been observed that the ground water sources have been decreased by 15% in the State during the Fourth Minor Irrigation Census as compared to the previous Census. North Goa district shows decrease of 22%, while South Goa represents 3% increase when compared to the previous Census. The cause for decrease in Ground water sources in North Goa district is mainly attributable to decrease in dugwells from 3748 to 2935 during the census in discussion. The main reason for increase of ground water sources in South Goa district was due to increase of Shallow Tube Wells from 16 to 70 and Deep Tube Well from 3 to 22.

5.3 **Surface Irrigation Sources:** The total number of surface irrigation sources recorded in the state during the current census is 2651 contributed by 1194 and 1457 in North Goa and South Goa Districts respectively. The number of surface irrigation schemes in the state has decreased by 45% as compared to the previous census. It is observed that the number of surface irrigation schemes have decreased by 63% in North

Goa District and 9% in South Goa District. The table below provides a comparative picture of the number of ground water and surface irrigation sources.

5.4 **Irrigated Area:** In the table below the total area irrigated through various minor irrigation sources is 11853 ha. as against 14125 ha. in the previous census, showing a remarkable decrease of 16.08% of the total irrigated area. Consequent upon decrease of 29.62% in the number of irrigation sources, a substantial decrease in irrigated area by 16.08% has been observed. This decrease is resulted largely owing to short listing of sources like Dug wells, Surface flow etc. by 16.52% & 54.91% respectively. Out of the total irrigated area 49 % and 51% are contributed by North and South Goa districts respectively. A decrease of 25.96% and 3.88% in irrigated areas in North and South Goa districts respectively has been observed. The decrease in area irrigated in North Goa District is mainly due to less contribution of Surface Flow Irrigation (50.90%) and Surface lift irrigation (10.59%). In South Goa district there is decrease of 39.65 % in Surface flow irrigation followed by decrease in Surface lift irrigation by (13.31%).

**TABLE – 5.2**

COMPARATIVE STATEMENT OF AREA IRRIGATED UNDER DIFFERENT IRRIGATION SOURCES

| State/District | Census | Area irrigated under Ground Water Sources | Area irrigated under Surface Irrigation Sources | Total area irrigated under Irrigation Sources |
|---|---|---|---|---|
| **Goa State** | 1st Census | 1962.00 | 9959.57 | 11921.58 |
| | 2nd Census | 3517.41 | 13182.29 | 16699.70 |
| | 3rd Census | 3602 | 10523 | 14125 |
| | 4th Census | 4959 | 6894 | 11853 |
| | *Variation (%) | (+37.67) | (-34.49) | (-16.08) |
| **North Goa District** | 1st Census | 1217.53 | 5699.70 | 6917.23 |
| | 2nd Census | 2239.07 | 7489.64 | 9728.70 |
| | 3rd Census | 2171 | 5638 | 7809 |
| | 4th Census | 2392 | 3390 | 5782 |
| | *Variation (%) | (+10.18) | (-39.87) | (-25.96) |
| **South Goa District** | 1st Census | 744.48 | 4259.87 | 5004.35 |
| | 2nd Census | 1278.35 | 5692.65 | 6971.00 |
| | 3rd Census | 1431 | 4885 | 6316 |
| | 4th Census | 2567 | 3504 | 6071 |
| | *Variation (%) | (+79.39) | (-28.27) | (-3.88) |

*Variation is as compared to 3rd Minor Irrigation Census

A comparative statement of taluka wise irrigation sources is presented in the Annexure–B

5.5 **Area Irrigated through Ground Water Sources:** The total irrigated area recorded under Ground water sources in the 4th Census of Minor irrigation is 4959 ha.(41.84% of total irrigated area) of which North and South Goa districts contributed 48.2% and 51.8% respectively, showing an increase of 37.67% over the previous census. It is observed that North and South Goa district showed an increase of 10.18% and 79.39% in its irrigated areas. Though the percentage of Ground water sources in the Census is 62.52%, the contribution of irrigated area under this sources is only 41.84%. Under this there is an increase of Area irrigated in North Goa and decrease in South Goa. The major area irrigated under Ground water of minor irrigation sources was mainly due to dug wells. The number of ground water sources has decreased by 21.57% but the area irrigated under this source has increased by 10.18% in North Goa district, while in South Goa District the number of Ground Water sources and also the area irrigated by this source have increased by 3% and 79.39% respectively.

5.6 **Irrigated Area under Surface Irrigation Schemes:** The total irrigated area under Surface irrigation schemes in the State is 6894 ha. contributing 58.16% of the total irrigated area. There is a decrease of 34.49% in the irrigated area under the scheme as compared to the previous census. Average decrease under irrigation is 39.87% and 28.27% in North and South Goa districts respectively. In terms of percentages, the contribution of irrigated area under the above source in North and South Goa districts are 49.17% and 50.83% respectively. There is a decrease of 63% in the number of surface irrigation schemes in North Goa district, the area covered by these sources for irrigation shows decrease of 39.87%. The picture seen in south Goa District is also similar as there is a decrease of 9.30% in the number of surface irrigation schemes and a decrease of 28.27% in the irrigated area.

# ANNEXURE - A

# &

# ANNEXURE - B

## ANNEXURE - A

## Comparative Statement of Taluka wise Irrigation Sources

(In Nos.)

| Sl. No. | Name of the Taluka/ District/ State | Dug Wells | | | | Shallow Tube Wells | | | | Deep Tube Wells | | | | Total (Ground Sources) | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 1st Census | 2nd Census | 3rd Census | 4th Census | 1st Census | 2nd Census | 3rd Census | 4th Census | 1st Census | 2nd Census | 3rd Census | 4th Census | 1st Census | 2nd Census | 3rd Census | 4th Census |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 |
| 1 | Bardez | 998 | 659 | 972 | 555 | 2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1000 | 659 | 972 | 555 |
| 2 | Bicholim | 333 | 330 | 509 | 640 | 0 | 0 | 0 | 2 | 0 | 0 | 0 | 0 | 333 | 330 | 509 | 642 |
| 3 | Pernem | 623 | 586 | 345 | 180 | 1 | 0 | 0 | 3 | 0 | 0 | 0 | 2 | 624 | 586 | 345 | 185 |
| 4 | Ponda | 420 | 502 | 649 | 679 | 0 | 0 | 5 | 14 | 0 | 2 | 0 | 7 | 420 | 504 | 654 | 700 |
| 5 | Sattari | 126 | 279 | 373 | 320 | 0 | 20 | 9 | 15 | 0 | 16 | 57 | 16 | 126 | 315 | 439 | 351 |
| 6 | Tiswadi | 1630 | 497 | 900 | 561 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1630 | 497 | 900 | 562 |
| **North Goa District** | | **4130** | **2853** | **3748** | **2935** | **3** | **20** | **14** | **35** | **0** | **18** | **57** | **25** | **4133** | **2891** | **3819** | **2995** |
| 7 | Canacona | 430 | 451 | 528 | 304 | 1 | 4 | 9 | 59 | 0 | 0 | 2 | 18 | 431 | 455 | 539 | 381 |
| 8 | Mormugao | 81 | 81 | 109 | 110 | 2 | 2 | 1 | 3 | 0 | 0 | 0 | 4 | 83 | 83 | 110 | 117 |
| 9 | Quepem | 93 | 114 | 178 | 260 | 0 | 0 | 1 | 1 | 2 | 0 | 0 | 0 | 95 | 114 | 179 | 261 |
| 10 | Salcete | 328 | 114 | 138 | 160 | 0 | 0 | 0 | 4 | 0 | 0 | 0 | 0 | 328 | 114 | 138 | 164 |
| 11 | Sanguem | 142 | 362 | 415 | 502 | 0 | 2 | 5 | 3 | 0 | 1 | 1 | 0 | 142 | 365 | 421 | 505 |
| **South Goa District** | | **1074** | **1122** | **1368** | **1336** | **3** | **8** | **16** | **70** | **2** | **1** | **3** | **22** | **1079** | **1131** | **1387** | **1428** |
| **Goa State** | | **5204** | **3975** | **5116** | **4271** | **6** | **28** | **30** | **105** | **2** | **19** | **60** | **47** | **5212** | **4022** | **5206** | **4423** |

(Contd….)

(Contd….)

(In Nos.)

| Sl. No. | Name of the Taluka/ District/ State | Surface Flow Irrigation | | | | Surface Lift Irrigation | | | | Total (Surface Sources) | | | | Total (All Sources) | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 1st Census | 2nd Census | 3rd Census | 4th Census | 1st Census | 2nd Census | 3rd Census | 4th Census | 1st Census | 2nd Census | 3rd Census | 4th Census | 1st Census | 2nd Census | 3rd Census | 4th Census |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 |
| 1 | Bardez | 139 | 680 | 1248 | 55 | 22 | 0 | 0 | 10 | 161 | 680 | 1248 | 65 | 1161 | 1339 | 2220 | 620 |
| 2 | Bicholim | 134 | 209 | 117 | 46 | 73 | 73 | 55 | 90 | 207 | 282 | 172 | 136 | 540 | 612 | 681 | 778 |
| 3 | Pernem | 200 | 169 | 164 | 34 | 64 | 25 | 78 | 67 | 264 | 194 | 242 | 101 | 888 | 780 | 587 | 286 |
| 4 | Ponda | 630 | 1043 | 809 | 612 | 29 | 37 | 79 | 79 | 659 | 1080 | 888 | 691 | 1079 | 1584 | 1542 | 1391 |
| 5 | Sattari | 354 | 179 | 242 | 95 | 111 | 154 | 207 | 94 | 465 | 333 | 449 | 189 | 591 | 648 | 888 | 540 |
| 6 | Tiswadi | 30 | 162 | 236 | 6 | 0 | 13 | 3 | 6 | 30 | 175 | 239 | 12 | 1660 | 672 | 1139 | 574 |
| **North Goa District** | | **1487** | **2442** | **2816** | **848** | **299** | **302** | **422** | **346** | **1786** | **2744** | **3238** | **1194** | **5919** | **5635** | **7057** | **4189** |
| | | | | | | | | | | | | | | | | | |
| 7 | Canacona | 705 | 289 | 194 | 239 | 101 | 105 | 24 | 39 | 806 | 394 | 218 | 278 | 1237 | 849 | 757 | 659 |
| 8 | Mormugao | 13 | 13 | 73 | 30 | 0 | 20 | 1 | 8 | 13 | 33 | 74 | 38 | 96 | 116 | 184 | 155 |
| 9 | Quepem | 266 | 88 | 248 | 195 | 61 | 78 | 101 | 86 | 327 | 166 | 349 | 281 | 422 | 280 | 528 | 542 |
| 10 | Salcete | 238 | 103 | 212 | 81 | 10 | 9 | 40 | 13 | 248 | 112 | 252 | 94 | 576 | 226 | 390 | 258 |
| 11 | Sanguem | 406 | 241 | 318 | 348 | 178 | 238 | 396 | 418 | 584 | 479 | 714 | 766 | 726 | 844 | 1135 | 1271 |
| **South Goa District** | | **1628** | **734** | **1045** | **893** | **350** | **450** | **562** | **564** | **1978** | **1184** | **1607** | **1457** | **3057** | **2315** | **2994** | **2885** |
| **Goa State** | | **3115** | **3176** | **3861** | **1741** | **649** | **752** | **984** | **910** | **3764** | **3928** | **4845** | **2651** | **8976** | **7950** | **10051** | **7074** |

## ANNEXURE - B

## Comparative Statement of Taluka wise net area irrigated under various Minor Irrigation Schemes

(In Ha.)

| Sl. No. | Name of the Taluka/ District/ State | Dug Wells | | | | Shallow Tube Wells | | | | Deep Tube Wells | | | | Total (Ground Sources) | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 1st Census | 2nd Census | 3rd Census | 4th Census | 1st Census | 2nd Census | 3rd Census | 4th Census | 1st Census | 2nd Census | 3rd Census | 4th Census | 1st Census | 2nd Census | 3rd Census | 4th Census |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 |
| 1 | Bardez | 116.00 | 131.90 | 255.00 | 331.00 | 0.70 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 116.7 | 131.9 | 255.00 | 331.00 |
| 2 | Bicholim | 136.83 | 316.85 | 280.00 | 366.00 | 0.00 | 0.00 | 0.00 | 1.00 | 0.00 | 0.00 | 0.00 | 0.00 | 136.83 | 316.85 | 280.00 | 367.00 |
| 3 | Pernem | 196.49 | 519.10 | 198.00 | 242.00 | 0.30 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 1.00 | 196.79 | 519.1 | 198.00 | 243.00 |
| 4 | Ponda | 252.96 | 591.48 | 539.00 | 432.00 | 0.00 | 0.00 | 10.00 | 18.00 | 0.00 | 10.50 | 0.00 | 11.00 | 252.96 | 601.98 | 549.00 | 461.00 |
| 5 | Sattari | 241.17 | 440.72 | 401.00 | 578.00 | 0.00 | 27.10 | 14.00 | 24.00 | 0.00 | 19.40 | 51.00 | 161.00 | 241.17 | 487.22 | 466.00 | 763.00 |
| 6 | Tiswadi | 273.08 | 182.01 | 423.00 | 227.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 273.08 | 182.01 | 423.00 | 227.00 |
| **North Goa District** | | **1216.53** | **2182.06** | **2096.00** | **2176.00** | **1.00** | **27.10** | **24.00** | **43.00** | **0.00** | **29.90** | **51.00** | **173.00** | **1217.53** | **2239.06** | **2171.00** | **2392.00** |
| 7 | Canacona | 171.71 | 236.20 | 264.00 | 782.00 | 2.00 | 1.55 | 10.00 | 49.00 | 0.00 | 0.00 | 3.00 | 13.00 | 173.71 | 237.75 | 277.00 | 844.00 |
| 8 | Mormugao | 23.76 | 44.63 | 23.00 | 55.00 | 0.90 | 1.00 | 1.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 24.66 | 45.63 | 24.00 | 55.00 |
| 9 | Quepem | 139.47 | 224.60 | 296.00 | 257.00 | 0.00 | 0.00 | 0.00 | 0.00 | 14.00 | 0.00 | 0.00 | 0.00 | 153.47 | 224.6 | 296.00 | 257.00 |
| 10 | Salcete | 101.64 | 75.66 | 114.00 | 36.00 | 0.00 | 0.00 | 0.00 | 11.00 | 0.00 | 0.00 | 0.00 | 0.00 | 101.64 | 75.66 | 114.00 | 47.00 |
| 11 | Sanguem | 291.00 | 674.01 | 689.00 | 1359.00 | 0.00 | 5.70 | 25.00 | 5.00 | 0.00 | 15.00 | 6.00 | 0.00 | 291 | 694.71 | 720.00 | 1364.00 |
| **South Goa District** | | **727.58** | **1255.10** | **1386.00** | **2489.00** | **2.90** | **8.25** | **36.00** | **65.00** | **14.00** | **15.00** | **9.00** | **13.00** | **744.48** | **1278.35** | **1431.00** | **2567.00** |
| **Goa State** | | **1944.11** | **3437.16** | **3482.00** | **4665.00** | **3.90** | **35.35** | **60.00** | **108.00** | **14.00** | **44.90** | **60.00** | **186.00** | **1962.01** | **3517.41** | **3602.00** | **4959.00** |

(Contd….)

(Contd….)

(In Ha.)

| Sl. No. | Name of the Taluka/ District/ State | Surface Flow Irrigation | | | | Surface Lift Irrigation | | | | Total (Surface Sources) | | | | Total (All Sources) | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 1st Census | 2nd Census | 3rd Census | 4th Census | 1st Census | 2nd Census | 3rd Census | 4th Census | 1st Census | 2nd Census | 3rd Census | 4th Census | 1st Census | 2nd Census | 3rd Census | 4th Census |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 |
| 1 | Bardez | 422.10 | 171.18 | 281.00 | 174.00 | 15.30 | 0.00 | 0.00 | 5.00 | 437.4 | 171.18 | 281.00 | 179.00 | 554.10 | 303.08 | 536.00 | 510.00 |
| 2 | Bicholim | 791.73 | 1736.12 | 786.00 | 413.00 | 96.10 | 135.00 | 152.00 | 291.00 | 887.83 | 1871.12 | 938.00 | 704.00 | 1024.66 | 2187.97 | 1218.00 | 1071.00 |
| 3 | Pernem | 1019.46 | 841.53 | 806.00 | 133.00 | 203.44 | 148.72 | 265.00 | 174.00 | 1222.9 | 990.25 | 1071.00 | 307.00 | 1419.69 | 1509.35 | 1269.00 | 550.00 |
| 4 | Ponda | 1633.49 | 2203.51 | 1500.00 | 1155.00 | 88.27 | 106.40 | 191.00 | 227.00 | 1721.76 | 2309.91 | 1691.00 | 1382.00 | 1974.72 | 2911.89 | 2240.00 | 1843.00 |
| 5 | Sattari | 731.22 | 448.96 | 641.00 | 171.00 | 568.45 | 815.28 | 815.00 | 577.00 | 1299.67 | 1264.24 | 1456.00 | 748.00 | 1540.84 | 1751.46 | 1922.00 | 1511.00 |
| 6 | Tiswadi | 130.14 | 874.39 | 197.00 | 66.00 | 0.00 | 8.55 | 4.00 | 4.00 | 130.14 | 882.94 | 201.00 | 70.00 | 403.22 | 1064.95 | 624.00 | 297.00 |
| **North Goa District** | | **4728.14** | **6275.69** | **4211.00** | **2112.00** | **971.56** | **1213.95** | **1427.00** | **1278.00** | **5699.7** | **7489.64** | **5638.00** | **3390.00** | **6917.23** | **9728.70** | **7809.00** | **5782.00** |
| | | | | | | | | | | | | | | | | | |
| 7 | Canacona | 250.11 | 335.46 | 432.00 | 463.00 | 67.85 | 94.10 | 35.00 | 55.00 | 317.96 | 429.56 | 467.00 | 518.00 | 491.67 | 667.31 | 744.00 | 1362.00 |
| 8 | Mormugao | 126.40 | 312.50 | 70.00 | 21.00 | 0.00 | 60.90 | 1.00 | 4.00 | 126.4 | 373.4 | 71.00 | 25.00 | 151.06 | 419.03 | 95.00 | 80.00 |
| 9 | Quepem | 331.27 | 299.50 | 306.00 | 357.00 | 276.65 | 650.80 | 360.00 | 486.00 | 607.92 | 950.3 | 666.00 | 843.00 | 761.39 | 1174.90 | 962.00 | 1100.00 |
| 10 | Salcete | 1095.02 | 873.90 | 811.00 | 73.00 | 4.22 | 10.90 | 48.00 | 5.00 | 1099.24 | 884.8 | 859.00 | 78.00 | 1200.88 | 960.46 | 973.00 | 125.00 |
| 11 | Sanguem | 1043.59 | 1382.90 | 1148.00 | 755.00 | 1064.76 | 1671.69 | 1674.00 | 1285.00 | 2108.35 | 3054.59 | 2822.00 | 2040.00 | 2399.35 | 3749.30 | 3542.00 | 3404.00 |
| **South Goa District** | | **2846.39** | **3204.26** | **2767.00** | **1669.00** | **1413.48** | **2488.39** | **2118.00** | **1835.00** | **4259.87** | **5692.65** | **4885.00** | **3504.00** | **5004.35** | **6971.00** | **6316.00** | **6071.00** |
| **Goa State** | | **7574.53** | **9479.95** | **6978.00** | **3781.00** | **2385.04** | **3702.34** | **3545.00** | **3113.00** | **9959.57** | **13182.3** | **10523.00** | **6894.00** | **11921.58** | **16699.70** | **14125.00** | **11853.00** |

# CHAPTER – VI

## SUMMARY OF IMPORTANT FINDINGS

i) The irrigation potential for Minor Irrigation Scheme under surface water is assessed at 82,260 ha. and as 7400 ha. under ground water sources. The irrigation potential so far created under surface water as well as ground water sources according to the data thrown up by Minor Irrigation Census is about 12,606 ha. There is, as such, considerable scope for increasing the irrigation potential under Minor Irrigation Schemes in Goa.

ii) In all 7074 schemes of Minor Irrigation were enumerated during the course of field operation, out of which 5761 or 81 percent were found to have been commissioned up to the year 2000-01. Dug wells formed the most common source of irrigation with 4271 schemes followed by Surface Flow Irrigation with 1741 schemes and Surface lift irrigation with 910 schemes. The number of tube wells and deep tube wells were 105 & 47 respectively.

iii) The total area irrigated through various minor irrigation sources decreased substantially as compared to previous Minor Irrigation Census. The irrigated area decreased from 14125 ha. in the previous census in 2000-01 to 11853 ha. showing a remarkable decrease of 16%. The decrease in irrigated area was observed in both the North as well as South Goa Districts.

iv) The Census has revealed that 93.5% of the Minor Irrigation schemes were in use.

v) Distribution of Dug wells and Shallow Tube wells according to operational holdings indicate that majority of the Dug wells i.e. 94.23 percent were located in the operational holdings of size class less than 1 ha. Out of 105 shallow tube wells in the state, 59 wells were in size class below 1 ha. and the remaining 46 wells in the size class of 1 ha. and above.

vi) The Culturable Command Area under Minor Irrigation Schemes is found to be 12198 ha. with Dug wells schemes accounting for 39.89% of the area.

vii) As against 12606 ha. irrigation potential created in the State, utilization or area irrigated formed (94.03%) thereby underlining the need for steps to be taken for further utilization of the potential available.

viii) Dug Wells Schemes were the most important sources irrigating almost 39.37% of the total irrigated area. The Surface Flow Irrigation Schemes irrigated about 31.88% followed by Surface Lift Schemes which irrigated 26.25% of the total area irrigated. The contribution of Shallow & Deep Tube wells was negligible.

# ANNEXURE – 1

# Fourth Census of Minor Irrigation Schemes 2006-07

# Statements

# 1. Dug Wells

## 1.1 Construction of Dug Wells over the years

(In Nos.)

| Sl. No. | Name of the State/District/Taluka | Construction of Dug Wells over the years | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | Up to 2000-01 | During 2001-02 | During 2002-03 | During 2003-04 | During 2004-05 | During 2005-06 | During 2006-07 | Total (3 to 9) |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 1 | Bardez | 362 | 127 | 21 | 12 | 4 | 13 | 16 | 555 |
| 2 | Bicholim | 572 | 19 | 12 | 18 | 7 | 5 | 7 | 640 |
| 3 | Pernem | 147 | 5 | 4 | 5 | 1 | 14 | 4 | 180 |
| 4 | Ponda | 537 | 47 | 33 | 22 | 11 | 20 | 9 | 679 |
| 5 | Sattari | 250 | 32 | 10 | 5 | 9 | 7 | 7 | 320 |
| 6 | Tiswadi | 544 | 4 | 1 | 4 | 2 | 4 | 2 | 561 |
| | **North Goa District** | **2412** | **234** | **81** | **66** | **34** | **63** | **45** | **2935** |
| 7 | Canacona | 243 | 10 | 7 | 7 | 11 | 4 | 22 | 304 |
| 8 | Mormugao | 39 | 4 | 0 | 1 | 0 | 1 | 65 | 110 |
| 9 | Quepem | 187 | 19 | 12 | 21 | 7 | 8 | 6 | 260 |
| 10 | Salcete | 110 | 5 | 5 | 3 | 33 | 0 | 4 | 160 |
| 11 | Sanguem | 355 | 141 | 2 | 2 | 0 | 2 | 0 | 502 |
| | **South Goa District** | **934** | **179** | **26** | **34** | **51** | **15** | **97** | **1336** |
| | **Goa State** | **3346** | **413** | **107** | **100** | **85** | **78** | **142** | **4271** |

## 1.2 Number of Dug Wells by Type

(In Nos.)

| Sl. No. | Name of the State/District/Taluka | Number of Dug Wells by Type | | | | |
|---|---|---|---|---|---|---|
| | | Pucca | Kucha | Dug cum Bore wells | Others | Total (3 to 6) |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| 1 | Bardez | 386 | 162 | 0 | 7 | 555 |
| 2 | Bicholim | 450 | 189 | 0 | 1 | 640 |
| 3 | Pernem | 141 | 34 | 0 | 5 | 180 |
| 4 | Ponda | 436 | 237 | 4 | 2 | 679 |
| 5 | Sattari | 258 | 60 | 2 | 0 | 320 |
| 6 | Tiswadi | 242 | 318 | 0 | 1 | 561 |
| | **North Goa District** | **1913** | **1000** | **6** | **16** | **2935** |
| 7 | Canacona | 263 | 33 | 6 | 2 | 304 |
| 8 | Mormugao | 42 | 67 | 0 | 1 | 110 |
| 9 | Quepem | 161 | 89 | 5 | 5 | 260 |
| 10 | Salcete | 22 | 136 | 1 | 1 | 160 |
| 11 | Sanguem | 387 | 115 | 0 | 0 | 502 |
| | **South Goa District** | **875** | **440** | **12** | **9** | **1336** |
| | **Goa State** | **2788** | **1440** | **18** | **25** | **4271** |

## 1.3 Number of Dug Wells by Owners Holding Size

(In Nos.)

| Sl. No. | Name of the State/District/Taluka | Number of Dug Wells with formers in Holding Group | | | | | |
|---|---|---|---|---|---|---|---|
| | | Marginal 0-1 | Small 1-2 | Semi Medium 2-4 | Medium 4-10 | Large (Big) >=10 | Total (3 to 7 ) |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| 1 | Bardez | 523 | 19 | 8 | 4 | 1 | 555 |
| 2 | Bicholim | 524 | 65 | 31 | 17 | 3 | 640 |
| 3 | Pernem | 149 | 17 | 8 | 3 | 3 | 180 |
| 4 | Ponda | 545 | 74 | 44 | 13 | 3 | 679 |
| 5 | Sattari | 106 | 74 | 69 | 48 | 23 | 320 |
| 6 | Tiswadi | 523 | 34 | 3 | 0 | 1 | 561 |
| | **North Goa District** | **2370** | **283** | **163** | **85** | **34** | **2935** |
| 7 | Canacona | 257 | 11 | 30 | 3 | 3 | 304 |
| 8 | Mormugao | 93 | 9 | 6 | 2 | 0 | 110 |
| 9 | Quepem | 166 | 37 | 32 | 17 | 8 | 260 |
| 10 | Salcete | 151 | 3 | 3 | 3 | 0 | 160 |
| 11 | Sanguem | 94 | 206 | 172 | 9 | 21 | 502 |
| | **South Goa District** | **761** | **266** | **243** | **34** | **32** | **1336** |
| | **Goa State** | **3131** | **549** | **406** | **119** | **66** | **4271** |

## 1.4 Number of Dug Wells according to the Social Status

(In Nos.)

| Sl. No. | Name of the State/District/Taluka | Number of Dug Wells belong to | | | | | Owned by other than individual/ group of farmers | Grand Total (7 +8) |
|---|---|---|---|---|---|---|---|---|
| | | SC | ST | Others | OBC | Total (3 to 6) | | |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| 1 | Bardez | 4 | 3 | 232 | 194 | 433 | 122 | 555 |
| 2 | Bicholim | 21 | 14 | 528 | 66 | 629 | 11 | 640 |
| 3 | Pernem | 1 | 0 | 87 | 39 | 127 | 53 | 180 |
| 4 | Ponda | 5 | 165 | 312 | 164 | 646 | 33 | 679 |
| 5 | Sattari | 3 | 4 | 256 | 2 | 265 | 55 | 320 |
| 6 | Tiswadi | 3 | 151 | 238 | 147 | 539 | 22 | 561 |
| | **North Goa District** | **37** | **337** | **1653** | **612** | **2027** | **296** | **2935** |
| 7 | Canacona | 0 | 31 | 186 | 21 | 238 | 66 | 304 |
| 8 | Mormugao | 0 | 3 | 37 | 31 | 71 | 39 | 110 |
| 9 | Quepem | 2 | 62 | 159 | 13 | 236 | 24 | 260 |
| 10 | Salcete | 0 | 43 | 88 | 9 | 140 | 20 | 160 |
| 11 | Sanguem | 61 | 265 | 159 | 12 | 497 | 5 | 502 |
| | **South Goa District** | **63** | **404** | **629** | **86** | **1096** | **154** | **1336** |
| | **Goa State** | **100** | **741** | **2282** | **698** | **3123** | **450** | **4271** |

## 1.5 Distribution of Dug wells according to Source of Energy

(In Nos.)

| Sl. No. | Name of the State/District/Taluka | No of Dug wells with Lifting Devices | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | Electric Pump | Diesel | Wind mills | Solar Pumps | Man / Animal Operated | Others | Total (3 to 8 ) |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| 1 | Bardez | 142 | 10 | 0 | 1 | 377 | 25 | 555 |
| 2 | Bicholim | 399 | 26 | 1 | 0 | 190 | 24 | 640 |
| 3 | Pernem | 149 | 8 | 0 | 0 | 18 | 5 | 180 |
| 4 | Ponda | 616 | 8 | 1 | 0 | 46 | 8 | 679 |
| 5 | Sattari | 237 | 6 | 0 | 0 | 2 | 75 | 320 |
| 6 | Tiswadi | 139 | 10 | 0 | 0 | 268 | 144 | 561 |
| | **North Goa District** | **1682** | **68** | **2** | **1** | **901** | **281** | **2935** |
| 7 | Canacona | 247 | 9 | 1 | 0 | 8 | 39 | 304 |
| 8 | Mormugao | 38 | 2 | 0 | 0 | 69 | 1 | 110 |
| 9 | Quepem | 176 | 25 | 0 | 0 | 15 | 44 | 260 |
| 10 | Salcete | 11 | 6 | 0 | 0 | 135 | 8 | 160 |
| 11 | Sanguem | 463 | 6 | 0 | 0 | 26 | 7 | 502 |
| | **South Goa District** | **935** | **48** | **1** | **0** | **253** | **99** | **1336** |
| | **Goa State** | **2617** | **116** | **3** | **1** | **1154** | **380** | **4271** |

## 1.6 Culturable Command Area and gross Irrigation Potential Created & Utilized by Dug Wells

(In ha.)

| Sl. No. | Name of the State/District/Taluka | No of Dug wells | | Culturable Command Area | Irrigation potential created by Dug wells | | | | | Gross Irrigation potential utilised by Dug Wells | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | In Use | Not in use | | Kharif | Rabi | Perinnial | Others | Total (6 to 9) | Kharif | Rabi | Perinnial | Others | Total (11 to 14) |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
| 1 | Bardez | 538 | 17 | 353 | 36 | 238 | 79 | 3 | 356 | 29 | 221 | 79 | 2 | 331 |
| 2 | Bicholim | 628 | 12 | 372 | 0 | 9 | 362 | 1 | 372 | 0 | 9 | 356 | 1 | 366 |
| 3 | Pernem | 175 | 5 | 234 | 52 | 77 | 128 | 6 | 263 | 51 | 62 | 123 | 6 | 242 |
| 4 | Ponda | 672 | 7 | 425 | 18 | 35 | 370 | 15 | 438 | 18 | 34 | 365 | 15 | 432 |
| 5 | Sattari | 246 | 74 | 717 | 1 | 119 | 571 | 27 | 718 | 1 | 93 | 462 | 22 | 578 |
| 6 | Tiswadi | 432 | 129 | 255 | 27 | 235 | 8 | 0 | 270 | 26 | 194 | 7 | 0 | 227 |
| | **North Goa District** | **2691** | **244** | **2356** | **134** | **713** | **1518** | **52** | **2417** | **125** | **613** | **1392** | **46** | **2176** |
| 7 | Canacona | 267 | 37 | 800 | 30 | 80 | 95 | 626 | 831 | 16 | 47 | 93 | 626 | 782 |
| 8 | Mormugao | 109 | 1 | 53 | 3 | 51 | 1 | 0 | 55 | 3 | 51 | 1 | 0 | 55 |
| 9 | Quepem | 236 | 24 | 277 | 1 | 71 | 200 | 5 | 277 | 1 | 53 | 198 | 5 | 257 |
| 10 | Salcete | 156 | 4 | 38 | 31 | 31 | 0 | 0 | 62 | 7 | 29 | 0 | 0 | 36 |
| 11 | Sanguem | 495 | 7 | 1342 | 10 | 249 | 1083 | 23 | 1365 | 10 | 246 | 1080 | 23 | 1359 |
| | **South Goa District** | **1263** | **73** | **2510** | **75** | **482** | **1379** | **654** | **2590** | **37** | **426** | **1372** | **654** | **2489** |
| | **Goa State** | **3954** | **317** | **4866** | **209** | **1195** | **2897** | **706** | **5007** | **162** | **1039** | **2764** | **700** | **4665** |

## 1.7 Area Irrigated from other sources but within the command area of Dug Wells

(In ha.)

| Sl. No. | Name of the State/District/Taluka | No of Dug Wells | Culturable Command Area | Net Area Irrigated from other sources but within the command area of Dug Wells | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | Kharif | Rabi | Perinnial | Others | Total (5 to 8) |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| 1 | Bardez | 555 | 353 | 0 | 0 | 0 | 0 | 0 |
| 2 | Bicholim | 640 | 372 | 0 | 0 | 18 | 0 | 18 |
| 3 | Pernem | 180 | 234 | 0 | 6 | 11 | 1 | 18 |
| 4 | Ponda | 679 | 425 | 0 | 0 | 0 | 0 | 0 |
| 5 | Sattari | 320 | 717 | 0 | 0 | 0 | 0 | 0 |
| 6 | Tiswadi | 561 | 255 | 0 | 0 | 0 | 0 | 0 |
| | **North Goa District** | **2935** | **2356** | **0** | **6** | **29** | **1** | **36** |
| 7 | Canacona | 304 | 800 | 0 | 0 | 0 | 0 | 0 |
| 8 | Mormugao | 110 | 53 | 0 | 0 | 0 | 0 | 0 |
| 9 | Quepem | 260 | 277 | 0 | 4 | 21 | 0 | 25 |
| 10 | Salcete | 160 | 38 | 0 | 0 | 0 | 0 | 0 |
| 11 | Sanguem | 502 | 1342 | 0 | 0 | 0 | 0 | 0 |
| | **South Goa District** | **1336** | **2510** | **0** | **4** | **21** | **0** | **25** |
| | **Goa State** | **4271** | **4866** | **0** | **10** | **50** | **1** | **61** |

## 1.8 No of Dug Wells according to the Constraints

(In Nos.)

| Sl. No. | Name of the State/District/Taluka | No of dug wells in use | No of dug wells with no constraints | Constraints in utilisation of Irrigation potential created | | | | | Grand Total (4 + 9) |
|---|---|---|---|---|---|---|---|---|---|
| | | | | Inadequate powers | Mechanical Breakdown | Less water discharge | Other Reasons | Total (5 to 8) | |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 1 | Bardez | 538 | 554 | 0 | 0 | 0 | 1 | 1 | 555 |
| 2 | Bicholim | 628 | 636 | 1 | 0 | 0 | 3 | 4 | 640 |
| 3 | Pernem | 175 | 175 | 0 | 0 | 0 | 5 | 5 | 180 |
| 4 | Ponda | 672 | 677 | 0 | 0 | 0 | 2 | 2 | 679 |
| 5 | Sattari | 246 | 304 | 4 | 1 | 0 | 11 | 16 | 320 |
| 6 | Tiswadi | 432 | 467 | 4 | 1 | 1 | 88 | 94 | 561 |
| | **North Goa District** | **2691** | **2813** | **9** | **2** | **1** | **110** | **122** | **2935** |
| 7 | Canacona | 267 | 268 | 0 | 11 | 7 | 18 | 36 | 304 |
| 8 | Mormugao | 109 | 109 | 0 | 0 | 0 | 1 | 1 | 110 |
| 9 | Quepem | 236 | 245 | 0 | 0 | 0 | 15 | 15 | 260 |
| 10 | Salcete | 156 | 40 | 0 | 0 | 1 | 119 | 120 | 160 |
| 11 | Sanguem | 495 | 502 | 0 | 0 | 0 | 0 | 0 | 502 |
| | **South Goa District** | **1263** | **1164** | **0** | **11** | **8** | **153** | **172** | **1336** |
| | **Goa State** | **3954** | **3977** | **9** | **13** | **9** | **263** | **294** | **4271** |

## 1.9 Number of Dug wells according to Water Distribution System

(In Nos.)

| Sl. No. | Name of State/District/Taluka | Sprinklers | Drip Irrigation | Open Channel | Surface pipe | Under Gr. Channel | Others | Total (3 to 8) |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| 1 | Bardez | 3 | 0 | 369 | 56 | 0 | 127 | 555 |
| 2 | Bicholim | 104 | 7 | 202 | 60 | 3 | 264 | 640 |
| 3 | Pernem | 10 | 0 | 126 | 23 | 0 | 21 | 180 |
| 4 | Ponda | 358 | 9 | 221 | 47 | 2 | 42 | 679 |
| 5 | Sattari | 72 | 2 | 129 | 34 | 4 | 79 | 320 |
| 6 | Tiswadi | 5 | 1 | 85 | 105 | 0 | 365 | 561 |
| | **North Goa District** | **552** | **19** | **1132** | **325** | **9** | **898** | **2935** |
| 7 | Canacona | 169 | 0 | 30 | 41 | 2 | 62 | 304 |
| 8 | Mormugao | 2 | 2 | 10 | 29 | 0 | 67 | 110 |
| 9 | Quepem | 31 | 22 | 119 | 45 | 9 | 34 | 260 |
| 10 | Salcete | 3 | 1 | 68 | 4 | 0 | 84 | 160 |
| 11 | Sanguem | 15 | 2 | 465 | 7 | 2 | 11 | 502 |
| | **South Goa District** | **220** | **27** | **692** | **126** | **13** | **258** | **1336** |
| | **Goa State** | **772** | **46** | **1824** | **451** | **22** | **1156** | **4271** |

## 1.10 Distribution of Dug wells according to Ownership

(In Nos.)

| Sl. No. | Name of State/District/Taluka | Govt. owned | Coop. Societies | Panchayat | Group of farmers | Individual farmers | Others | Total (3 to 8) |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| 1 | Bardez | 46 | 0 | 1 | 38 | 433 | 37 | 555 |
| 2 | Bicholim | 4 | 0 | 0 | 7 | 629 | 0 | 640 |
| 3 | Pernem | 16 | 1 | 1 | 32 | 127 | 3 | 180 |
| 4 | Ponda | 3 | 0 | 0 | 28 | 646 | 2 | 679 |
| 5 | Sattari | 26 | 0 | 0 | 8 | 265 | 21 | 320 |
| 6 | Tiswadi | 21 | 0 | 0 | 1 | 539 | 0 | 561 |
| | **North Goa District** | **116** | **1** | **2** | **114** | **2639** | **63** | **2935** |
| 7 | Canacona | 22 | 0 | 3 | 11 | 238 | 30 | 304 |
| 8 | Mormugao | 0 | 0 | 0 | 1 | 71 | 38 | 110 |
| 9 | Quepem | 2 | 2 | 0 | 16 | 236 | 4 | 260 |
| 10 | Salcete | 0 | 0 | 0 | 16 | 140 | 4 | 160 |
| 11 | Sanguem | 0 | 4 | 0 | 1 | 497 | 0 | 502 |
| | **South Goa District** | **24** | **6** | **3** | **45** | **1182** | **76** | **1336** |
| | **Goa State** | **140** | **7** | **5** | **159** | **3821** | **139** | **4271** |

## 1.11 Distribution of Dug wells according the Source of Finance

(In Nos.)

| Sl. No. | Name of State/District/Taluka | Bank Loan | Govt. Finance | Own Saving | Money Lenders | Others | Total (3 to 7) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| 1 | Bardez | 38 | 149 | 317 | 8 | 43 | 555 |
| 2 | Bicholim | 41 | 16 | 559 | 1 | 23 | 640 |
| 3 | Pernem | 9 | 32 | 128 | 0 | 11 | 180 |
| 4 | Ponda | 101 | 42 | 517 | 1 | 18 | 679 |
| 5 | Sattari | 28 | 142 | 127 | 2 | 21 | 320 |
| 6 | Tiswadi | 97 | 36 | 397 | 1 | 30 | 561 |
| | **North Goa District** | **314** | **417** | **2045** | **13** | **146** | **2935** |
| 7 | Canacona | 28 | 37 | 205 | 0 | 34 | 304 |
| 8 | Mormugao | 10 | 2 | 94 | 1 | 3 | 110 |
| 9 | Quepem | 45 | 13 | 198 | 0 | 4 | 260 |
| 10 | Salcete | 1 | 1 | 153 | 0 | 5 | 160 |
| 11 | Sanguem | 422 | 22 | 58 | 0 | 0 | 502 |
| | **South Goa District** | **506** | **75** | **708** | **1** | **46** | **1336** |
| | **Goa State** | **820** | **492** | **2753** | **14** | **192** | **4271** |

# 2. SHALLOW TUBE WELLS

## 2.1 Construction of Shallow Tube Wells over the years

(In Nos.)

| Sl. No. | Name of the State/District/Taluka | Construction of Shallow Tube Wells over the years | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | Up to 2000-01 | During 2001-02 | During 2002-03 | During 2003-04 | During 2004-05 | During 2005-06 | During 2006-07 | Total (3 to 9) |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 1 | Bardez | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2 | Bicholim | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 2 |
| 3 | Pernem | 1 | 1 | 0 | 1 | 0 | 0 | 0 | 3 |
| 4 | Ponda | 10 | 1 | 3 | 0 | 0 | 0 | 0 | 14 |
| 5 | Sattari | 13 | 0 | 2 | 0 | 0 | 0 | 0 | 15 |
| 6 | Tiswadi | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
| | **North Goa District** | **26** | **2** | **6** | **1** | **0** | **0** | **0** | **35** |
| 7 | Canacona | 31 | 3 | 7 | 9 | 8 | 1 | 0 | 59 |
| 8 | Mormugao | 3 | 0 | 0 | 0 | 0 | 0 | 0 | 3 |
| 9 | Quepem | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
| 10 | Salcete | 4 | 0 | 0 | 0 | 0 | 0 | 0 | 4 |
| 11 | Sanguem | 3 | 0 | 0 | 0 | 0 | 0 | 0 | 3 |
| | **South Goa District** | **42** | **3** | **7** | **9** | **8** | **1** | **0** | **70** |
| | **Goa State** | **68** | **5** | **13** | **10** | **8** | **1** | **0** | **105** |

## 2.2 Number of Shallow Tube Wells by Type

(In Nos.)

| Sl. No. | Name of the State/District/Taluka | Type of Shallow Tube Wells | | | | Total (3 to 6) |
|---|---|---|---|---|---|---|
| | | Shallow | Filter | Bore wells | Others | |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| 1 | Bardez | 0 | 0 | 0 | 0 | 0 |
| 2 | Bicholim | 0 | 2 | 0 | 0 | 2 |
| 3 | Pernem | 1 | 2 | 0 | 0 | 3 |
| 4 | Ponda | 11 | 1 | 0 | 2 | 14 |
| 5 | Sattari | 15 | 0 | 0 | 0 | 15 |
| 6 | Tiswadi | 1 | 0 | 0 | 0 | 1 |
| | **North Goa District** | **28** | **5** | **0** | **2** | **35** |
| 7 | Canacona | 41 | 18 | 0 | 0 | 59 |
| 8 | Mormugao | 3 | 0 | 0 | 0 | 3 |
| 9 | Quepem | 0 | 1 | 0 | 0 | 1 |
| 10 | Salcete | 4 | 0 | 0 | 0 | 4 |
| 11 | Sanguem | 3 | 0 | 0 | 0 | 3 |
| | **South Goa District** | **51** | **19** | **0** | **0** | **70** |
| | **Goa State** | **79** | **24** | **0** | **2** | **105** |

## 2.3 Number of Shallow Tube Wells by Owners Holding Size

(In Nos.)

| Sl. No. | Name of the State/District/Taluka | Number of Shallow Tube Wells with farmers in Holding Group | | | | | |
|---|---|---|---|---|---|---|---|
| | | Marginal 0-1 | Small 1-2 | Semi Medium 2-4 | Medium 4-10 | Large (Big) >=10 | Total (3 to 7 ) |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| 1 | Bardez | 0 | 0 | 0 | 0 | 0 | 0 |
| 2 | Bicholim | 2 | 0 | 0 | 0 | 0 | 2 |
| 3 | Pernem | 3 | 0 | 0 | 0 | 0 | 3 |
| 4 | Ponda | 8 | 2 | 1 | 2 | 1 | 14 |
| 5 | Sattari | 6 | 3 | 5 | 1 | 0 | 15 |
| 6 | Tiswadi | 1 | 0 | 0 | 0 | 0 | 1 |
| | **North Goa District** | **20** | **5** | **6** | **3** | **1** | **35** |
| 7 | Canacona | 34 | 8 | 14 | 3 | 0 | 59 |
| 8 | Mormugao | 3 | 0 | 0 | 0 | 0 | 3 |
| 9 | Quepem | 1 | 0 | 0 | 0 | 0 | 1 |
| 10 | Salcete | 1 | 1 | 0 | 2 | 0 | 4 |
| 11 | Sanguem | 0 | 2 | 0 | 0 | 1 | 3 |
| | **South Goa District** | **39** | **11** | **14** | **5** | **1** | **70** |
| | **Goa State** | **59** | **16** | **20** | **8** | **2** | **105** |

## 2.4 Number of Shallow Tube Wells according to the Social Status

(In Nos.)

| Sl. No. | Name of the State/District/Taluka | Number of Shallow Tube Wells belong to | | | | | Owned by other than individual/ group of farmers | Grand Total (7 +8) |
|---|---|---|---|---|---|---|---|---|
| | | SC | ST | Others | OBC | Total (3 to 6) | | |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| 1 | Bardez | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2 | Bicholim | 0 | 0 | 2 | 0 | 2 | 0 | 2 |
| 3 | Pernem | 0 | 0 | 0 | 1 | 1 | 2 | 3 |
| 4 | Ponda | 0 | 3 | 8 | 2 | 13 | 1 | 14 |
| 5 | Sattari | 0 | 0 | 10 | 0 | 10 | 5 | 15 |
| 6 | Tiswadi | 0 | 0 | 1 | 0 | 1 | 0 | 1 |
| | **North Goa District** | **0** | **3** | **21** | **3** | **27** | **8** | **35** |
| 7 | Canacona | 0 | 24 | 3 | 0 | 27 | 32 | 59 |
| 8 | Mormugao | 0 | 0 | 1 | 2 | 3 | 0 | 3 |
| 9 | Quepem | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
| 10 | Salcete | 0 | 0 | 4 | 0 | 4 | 0 | 4 |
| 11 | Sanguem | 0 | 1 | 2 | 0 | 3 | 0 | 3 |
| | **South Goa District** | **0** | **25** | **10** | **2** | **37** | **33** | **70** |
| | **Goa State** | **0** | **28** | **31** | **5** | **64** | **41** | **105** |

## 2.5 Distribution of Shallow Tube Wells according to Source of Energy

(In Nos.)

| Sl. No. | Name of the State/District/Taluka | No of Shallow Tube Wells with Lifting Devices | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | Electric Pump | Diesel | Wind mills | Solar Pumps | Man / Animal Operated | Others | Total (3 to 8 ) |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| 1 | Bardez | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2 | Bicholim | 2 | 0 | 0 | 0 | 0 | 0 | 2 |
| 3 | Pernem | 1 | 0 | 0 | 0 | 1 | 1 | 3 |
| 4 | Ponda | 13 | 0 | 0 | 0 | 0 | 1 | 14 |
| 5 | Sattari | 12 | 0 | 0 | 0 | 0 | 3 | 15 |
| 6 | Tiswadi | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
| | **North Goa District** | **28** | **0** | **0** | **0** | **1** | **6** | **35** |
| 7 | Canacona | 25 | 15 | 1 | 0 | 3 | 15 | 59 |
| 8 | Mormugao | 3 | 0 | 0 | 0 | 0 | 0 | 3 |
| 9 | Quepem | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
| 10 | Salcete | 2 | 2 | 0 | 0 | 0 | 0 | 4 |
| 11 | Sanguem | 3 | 0 | 0 | 0 | 0 | 0 | 3 |
| | **South Goa District** | **33** | **17** | **1** | **0** | **3** | **16** | **70** |
| | **Goa State** | **61** | **17** | **1** | **0** | **4** | **22** | **105** |

## 2.6 Culturable Command Area and Gross Irrigation Potential Created & Utilised by Shallow Tube Wells

(In ha.)

| Sl. No. | Name of the State/District/Taluka | No of Shallow Tube Wells | | Culturable Command area | Irrigation potential created by Shallow Tube Wells | | | | | Irrigation potential utilised by Shallow Tube Wells | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | In Use | Not in Use | | Kharif | Rabi | Perennial | Others | Total (6 to 9) | Kharif | Rabi | Perinnial | Others | Total (11 to 14) |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
| 1 | Bardez | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2 | Bicholim | 2 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 1 |
| 3 | Pernem | 2 | 1 | 2 | 1 | 1 | 1 | 0 | 3 | 0 | 0 | 0 | 0 | 0 |
| 4 | Ponda | 13 | 1 | 22 | 0 | 0 | 22 | 0 | 22 | 0 | 0 | 18 | 0 | 18 |
| 5 | Sattari | 13 | 2 | 26 | 0 | 2 | 25 | 0 | 27 | 0 | 2 | 22 | 0 | 24 |
| 6 | Tiswadi | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | **North Goa District** | **30** | **5** | **51** | **1** | **3** | **49** | **0** | **53** | **0** | **2** | **41** | **0** | **43** |
| 7 | Canacona | 44 | 15 | 67 | 72 | 9 | 4 | 0 | 85 | 39 | 8 | 2 | 0 | 49 |
| 8 | Mormugao | 3 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 9 | Quepem | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 10 | Salcete | 4 | 0 | 11 | 10 | 11 | 0 | 0 | 21 | 0 | 11 | 0 | 0 | 11 |
| 11 | Sanguem | 3 | 0 | 5 | 0 | 0 | 5 | 0 | 5 | 0 | 0 | 5 | 0 | 5 |
| | **South Goa District** | **54** | **16** | **83** | **82** | **20** | **9** | **0** | **111** | **39** | **19** | **7** | **0** | **65** |
| | **Goa State** | **84** | **21** | **134** | **83** | **23** | **58** | **0** | **164** | **39** | **21** | **48** | **0** | **108** |

## 2.7 Area Irrigated from other sources but within the command area of Shallow Tube Wells

(In ha.)

| Sl. No. | Name of the State/District/Taluka | No of Shallow Tube Wells | Culturable Command Area | Net Area Irrigated from other sources but within the command area of Shallow Tube Wells | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | Kharif | Rabi | Perinnial | Others | Total (5 to 8) |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| 1 | Bardez | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2 | Bicholim | 2 | 1 | 0 | 0 | 0 | 0 | 0 |
| 3 | Pernem | 3 | 2 | 0 | 0 | 0 | 0 | 0 |
| 4 | Ponda | 14 | 22 | 0 | 0 | 0 | 0 | 0 |
| 5 | Sattari | 15 | 26 | 0 | 0 | 0 | 0 | 0 |
| 6 | Tiswadi | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| | **North Goa District** | **35** | **51** | **0** | **0** | **0** | **0** | **0** |
| 7 | Canacona | 59 | 67 | 0 | 0 | 0 | 0 | 0 |
| 8 | Mormugao | 3 | 0 | 0 | 0 | 0 | 0 | 0 |
| 9 | Quepem | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| 10 | Salcete | 4 | 11 | 0 | 0 | 0 | 0 | 0 |
| 11 | Sanguem | 3 | 5 | 0 | 0 | 0 | 0 | 0 |
| | **South Goa District** | **70** | **83** | **0** | **0** | **0** | **0** | **0** |
| | **Goa State** | **105** | **134** | **0** | **0** | **0** | **0** | **0** |

## 2.8 No of Shallow Tube Wells according to the Constraints

(In Nos.)

| Sl. No. | Name of the State/District/Taluka | No of Shallow Tube Wells in use | No of Shallow Tube Wells with no constraints | Constraints in utilisation of Irrigation potential created | | | | | Grand Total (4 + 9) |
|---|---|---|---|---|---|---|---|---|---|
| | | | | Inadequate powers | Mechanical Breakdown | Less water discharge | Other Reasons | Total (5 to 8) | |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 1 | Bardez | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2 | Bicholim | 2 | 2 | 0 | 0 | 0 | 0 | 0 | 2 |
| 3 | Pernem | 2 | 3 | 0 | 0 | 0 | 0 | 0 | 3 |
| 4 | Ponda | 13 | 10 | 0 | 0 | 0 | 4 | 4 | 14 |
| 5 | Sattari | 13 | 15 | 0 | 0 | 0 | 0 | 0 | 15 |
| 6 | Tiswadi | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 |
| | **North Goa District** | **30** | **31** | **0** | **0** | **0** | **4** | **4** | **35** |
| 7 | Canacona | 44 | 31 | 0 | 10 | 13 | 5 | 28 | 59 |
| 8 | Mormugao | 3 | 3 | 0 | 0 | 0 | 0 | 0 | 3 |
| 9 | Quepem | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 |
| 10 | Salcete | 4 | 2 | 0 | 0 | 0 | 2 | 2 | 4 |
| 11 | Sanguem | 3 | 3 | 0 | 0 | 0 | 0 | 0 | 3 |
| | **South Goa District** | **54** | **39** | **0** | **10** | **13** | **8** | **31** | **70** |
| | **Goa State** | **84** | **70** | **0** | **10** | **13** | **12** | **35** | **105** |

## 2.9 Number of Shallow Tube Wells according to Water Distribution System

(In Nos.)

| Sl. No. | Name of State/District/Taluka | Sprinklers | Drip Irrigation | Open Channel | Surface pipe | Under Gr. Channel | Others | Total (3 to 8) |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| 1 | Bardez | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2 | Bicholim | 0 | 0 | 1 | 0 | 0 | 1 | 2 |
| 3 | Pernem | 0 | 0 | 2 | 0 | 0 | 1 | 3 |
| 4 | Ponda | 8 | 0 | 1 | 4 | 0 | 1 | 14 |
| 5 | Sattari | 3 | 0 | 8 | 2 | 0 | 2 | 15 |
| 6 | Tiswadi | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
| | **North Goa District** | **11** | **0** | **12** | **6** | **0** | **6** | **35** |
| 7 | Canacona | 0 | 0 | 4 | 34 | 1 | 20 | 59 |
| 8 | Mormugao | 0 | 0 | 1 | 2 | 0 | 0 | 3 |
| 9 | Quepem | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
| 10 | Salcete | 2 | 0 | 0 | 1 | 0 | 1 | 4 |
| 11 | Sanguem | 0 | 0 | 3 | 0 | 0 | 0 | 3 |
| | **South Goa District** | **2** | **0** | **8** | **37** | **1** | **22** | **70** |
| | **Goa State** | **13** | **0** | **20** | **43** | **1** | **28** | **105** |

## 2.10 Distribution of Shallow Tube Wells according to Ownership

(In Nos.)

| Sl. No. | Name of State/District/Taluka | Govt. owned | Coop. Societies | Panchayat | Group of farmers | Individual farmers | Others | Total (3 to 8) |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| 1 | Bardez | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2 | Bicholim | 0 | 0 | 0 | 0 | 2 | 0 | 2 |
| 3 | Pernem | 0 | 0 | 0 | 1 | 1 | 1 | 3 |
| 4 | Ponda | 0 | 0 | 0 | 0 | 13 | 1 | 14 |
| 5 | Sattari | 4 | 0 | 0 | 1 | 10 | 0 | 15 |
| 6 | Tiswadi | 0 | 0 | 0 | 0 | 1 | 0 | 1 |
| | **North Goa District** | **4** | **0** | **0** | **2** | **27** | **2** | **35** |
| 7 | Canacona | 25 | 0 | 2 | 2 | 27 | 3 | 59 |
| 8 | Mormugao | 0 | 0 | 0 | 0 | 3 | 0 | 3 |
| 9 | Quepem | 0 | 1 | 0 | 0 | 0 | 0 | 1 |
| 10 | Salcete | 0 | 0 | 0 | 0 | 4 | 0 | 4 |
| 11 | Sanguem | 0 | 0 | 0 | 0 | 3 | 0 | 3 |
| | **South Goa District** | **25** | **1** | **2** | **2** | **37** | **3** | **70** |
| | **Goa State** | **29** | **1** | **2** | **4** | **64** | **5** | **105** |

## 2.11 Distribution of Shallow Tube Wells according the Source of Finance

(In Nos.)

| Sl. No. | Name of State/District/Taluka | Bank Loan | Govt. Finance | Own Saving | Money Lenders | Others | Total (3 to 7) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| 1 | Bardez | 0 | 0 | 0 | 0 | 0 | 0 |
| 2 | Bicholim | 1 | 0 | 1 | 0 | 0 | 2 |
| 3 | Pernem | 0 | 1 | 1 | 0 | 1 | 3 |
| 4 | Ponda | 0 | 8 | 5 | 0 | 1 | 14 |
| 5 | Sattari | 0 | 13 | 2 | 0 | 0 | 15 |
| 6 | Tiswadi | 0 | 0 | 1 | 0 | 0 | 1 |
| | **North Goa District** | **1** | **22** | **10** | **0** | **2** | 35 |
| 7 | Canacona | 0 | 22 | 30 | 0 | 7 | 59 |
| 8 | Mormugao | 0 | 0 | 3 | 0 | 0 | 3 |
| 9 | Quepem | 0 | 0 | 1 | 0 | 0 | 1 |
| 10 | Salcete | 0 | 0 | 3 | 0 | 1 | 4 |
| 11 | Sanguem | 2 | 1 | 0 | 0 | 0 | 3 |
| | **South Goa District** | **2** | **23** | **37** | **0** | **8** | 70 |
| | **Goa State** | **3** | **45** | **47** | **0** | **10** | **105** |

# 3. DEEP TUBE WELLS

## 3.1 Construction of Deep Tube Wells over the years

(In Nos.)

| Sl. No. | Name of the State/District/Taluka | Construction of Deep Tube over the years | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | Up to 2000-01 | During 2001-02 | During 2002-03 | During 2003-04 | During 2004-05 | During 2005-06 | During 2006-07 | Total (3 to 10) |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 1 | Bardez | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2 | Bicholim | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 3 | Pernem | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 2 |
| 4 | Ponda | 4 | 2 | 0 | 0 | 0 | 0 | 1 | 7 |
| 5 | Sattari | 10 | 5 | 0 | 1 | 0 | 0 | 0 | 16 |
| 6 | Tiswadi | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | **North Goa District** | **14** | **8** | **0** | **2** | **0** | **0** | **1** | **25** |
| 7 | Canacona | 13 | 2 | 1 | 1 | 1 | 0 | 0 | 18 |
| 8 | Mormugao | 4 | 0 | 0 | 0 | 0 | 0 | 0 | 4 |
| 9 | Quepem | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 10 | Salcete | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 11 | Sanguem | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | **South Goa District** | **17** | **2** | **1** | **1** | **1** | **0** | **0** | **22** |
| | **Goa State** | **31** | **10** | **1** | **3** | **1** | **0** | **1** | **47** |

## 3.2 Number of Deep Tube Wells by type

(In Nos.)

| Sl. No. | Name of the State/District/Taluka | Type of Deep Tube Wells | | Total (3+4) |
|---|---|---|---|---|
| | | Deep Tube well | Others | |
| 1 | 2 | 3 | 4 | 5 |
| 1 | Bardez | 0 | 0 | 0 |
| 2 | Bicholim | 0 | 0 | 0 |
| 3 | Pernem | 2 | 0 | 2 |
| 4 | Ponda | 5 | 2 | 7 |
| 5 | Sattari | 15 | 1 | 16 |
| 6 | Tiswadi | 0 | 0 | 0 |
| | **North Goa District** | **22** | **3** | **25** |
| 7 | Canacona | 7 | 11 | **18** |
| 8 | Mormugao | 4 | 0 | **4** |
| 9 | Quepem | 0 | 0 | **0** |
| 10 | Salcete | 0 | 0 | **0** |
| 11 | Sanguem | 0 | 0 | **0** |
| | **South Goa District** | **11** | **11** | **22** |
| | **Goa State** | **33** | **14** | **47** |

## 3.3 Number of Deep Tube Wells by Owners Holding size

(In Nos.)

| Sl. No. | Name of the State/District/Taluka | Number of Deep Tube Wells with farmers in Holding Group | | | | | |
|---|---|---|---|---|---|---|---|
| | | Marginal 0-1 | Small 1-2 | Semi Medium 2-4 | Medium 4-10 | Large (Big) >=10 | Total (3 to 7 ) |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| 1 | Bardez | 0 | 0 | 0 | 0 | 0 | 0 |
| 2 | Bicholim | 0 | 0 | 0 | 0 | 0 | 0 |
| 3 | Pernem | 2 | 0 | 0 | 0 | 0 | 2 |
| 4 | Ponda | 2 | 2 | 2 | 0 | 1 | 7 |
| 5 | Sattari | 2 | 3 | 1 | 3 | 7 | 16 |
| 6 | Tiswadi | 0 | 0 | 0 | 0 | 0 | 0 |
| | **North Goa District** | **6** | **5** | **3** | **3** | **8** | **25** |
| 7 | Canacona | 15 | 2 | 1 | 0 | 0 | 18 |
| 8 | Mormugao | 4 | 0 | 0 | 0 | 0 | 4 |
| 9 | Quepem | 0 | 0 | 0 | 0 | 0 | 0 |
| 10 | Salcete | 0 | 0 | 0 | 0 | 0 | 0 |
| 11 | Sanguem | 0 | 0 | 0 | 0 | 0 | 0 |
| | **South Goa District** | **19** | **2** | **1** | **0** | **0** | **22** |
| | **Goa State** | **25** | **7** | **4** | **3** | **8** | **47** |

## 3.4 Number of Deep Tube Wells according to the Social Status

(In Nos.)

| Sl. No. | Name of the State/District/Taluka | Number of Deep Tube Wells belong to | | | | | Owned by other than individual/ group of farmers | Grand Total (6 + 7) |
|---|---|---|---|---|---|---|---|---|
| | | SC | ST | Others | OBC | Total (3 to 6) | | |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| 1 | Bardez | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2 | Bicholim | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 3 | Pernem | 0 | 0 | 1 | 0 | 1 | 1 | 2 |
| 4 | Ponda | 0 | 2 | 3 | 0 | 5 | 2 | 7 |
| 5 | Sattari | 0 | 0 | 15 | 0 | 15 | 1 | 16 |
| 6 | Tiswadi | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | **North Goa District** | **0** | **2** | **19** | **0** | **21** | **4** | **25** |
| 7 | Canacona | 0 | 2 | 2 | 0 | 4 | 14 | 18 |
| 8 | Mormugao | 0 | 0 | 3 | 0 | 3 | 1 | 4 |
| 9 | Quepem | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 10 | Salcete | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 11 | Sanguem | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | **South Goa District** | **0** | **2** | **5** | **0** | **7** | **15** | **22** |
| | **Goa State** | **0** | **4** | **24** | **0** | **28** | **19** | **47** |

## 3.5 Number of Deep Tube Wells according to Lifting Devices

(In Nos.)

| Sl. No. | Name of the State/District/Taluka | No of Deep Tube Wells with Water Lifting Devices | | | | | |
|---|---|---|---|---|---|---|---|
| | | Submersible Pump | Centrifugal Pump | Turbine Pump | Manual /Animal | Others | Total (3 to 7 ) |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| 1 | Bardez | 0 | 0 | 0 | 0 | 0 | 0 |
| 2 | Bicholim | 0 | 0 | 0 | 0 | 0 | 0 |
| 3 | Pernem | 0 | 1 | 0 | 1 | 0 | 2 |
| 4 | Ponda | 2 | 5 | 0 | 0 | 0 | 7 |
| 5 | Sattari | 14 | 2 | 0 | 0 | 0 | 16 |
| 6 | Tiswadi | 0 | 0 | 0 | 0 | 0 | 0 |
| | **North Goa District** | **16** | **8** | **0** | **1** | **0** | 25 |
| 7 | Canacona | 8 | 3 | 0 | 0 | 7 | 18 |
| 8 | Mormugao | 0 | 0 | 0 | 0 | 4 | 4 |
| 9 | Quepem | 0 | 0 | 0 | 0 | 0 | 0 |
| 10 | Salcete | 0 | 0 | 0 | 0 | 0 | 0 |
| 11 | Sanguem | 0 | 0 | 0 | 0 | 0 | 0 |
| | **South Goa District** | **8** | **3** | **0** | **0** | **11** | **22** |
| | **Goa State** | **24** | **11** | **0** | **1** | **11** | **47** |

## 3.6 Culturable Command Area and Gross Irrigation Potential Created & Utilised by Deep Tube Wells

(In ha.)

| Sl. No. | Name of the State/District/Taluka | No of Deep Tube Wells | | Culturable Command area | Irrigation potential created by Deep Tube Wells | | | | | Irrigation potential utilised by Deep Tube Wells | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | In Use | Not in Use | | Kharif | Rabi | Perinnial | Others | Total (6 to 9) | Kharif | Rabi | Perinnial | Others | Total (11 to 14) |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
| 1 | Bardez | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2 | Bicholim | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 3 | Pernem | 2 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 1 |
| 4 | Ponda | 7 | 0 | 10 | 2 | 1 | 6 | 2 | 11 | 2 | 1 | 6 | 2 | 11 |
| 5 | Sattari | 16 | 0 | 162 | 0 | 76 | 63 | 22 | 161 | 0 | 76 | 63 | 22 | 161 |
| 6 | Tiswadi | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | **North Goa District** | **25** | **0** | **173** | **2** | **78** | **69** | **24** | **173** | **2** | **78** | **69** | **24** | **173** |
| 7 | Canacona | 13 | 5 | 21 | 8 | 19 | 0 | 0 | 27 | 3 | 10 | 0 | 0 | 13 |
| 8 | Mormugao | 0 | 4 | 3 | 0 | 0 | 3 | 0 | 3 | 0 | 0 | 0 | 0 | 0 |
| 9 | Quepem | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 10 | Salcete | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 11 | Sanguem | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | **South Goa District** | **13** | **9** | **24** | **8** | **19** | **3** | **0** | **30** | **3** | **10** | **0** | **0** | **13** |
| | **Goa State** | **38** | **9** | **197** | **10** | **97** | **72** | **24** | **203** | **5** | **88** | **69** | **24** | **186** |

## 3.7 Area Irrigated from other sources but within the command area of Deep Tube Wells

(In ha.)

| Sl. No. | Name of the State/District/Taluka | No of Deep Tube Wells | Culturable Command area | Net Area Irrigated from other sources but within the command area of Deep Tube Wells | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | Kharif | Rabi | Perinnial | Others | Total (5 to 8) |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| 1 | Bardez | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2 | Bicholim | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 3 | Pernem | 2 | 1 | 0 | 0 | 0 | 0 | 0 |
| 4 | Ponda | 7 | 10 | 0 | 0 | 0 | 0 | 0 |
| 5 | Sattari | 16 | 162 | 0 | 0 | 0 | 0 | 0 |
| 6 | Tiswadi | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | **North Goa District** | **25** | **173** | **0** | **0** | **0** | **0** | **0** |
| 7 | Canacona | 18 | 21 | 0 | 0 | 0 | 0 | 0 |
| 8 | Mormugao | 4 | 3 | 0 | 0 | 0 | 0 | 0 |
| 9 | Quepem | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 10 | Salcete | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 11 | Sanguem | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | **South Goa District** | **22** | **24** | **0** | **0** | **0** | **0** | **0** |
| | **Goa State** | **47** | **197** | **0** | **0** | **0** | **0** | **0** |

## 3.8 No of Deep Tube Wells according to the Constraints

(In Nos.)

| Sl. No. | Name of the State/District/Taluka | No of Deep Tube Wells in use | No of Deep Tube Wells with no constraints | Constraints in utilisation of Irrigation potential created | | | | | Grand Total (4 + 9) |
|---|---|---|---|---|---|---|---|---|---|
| | | | | Inadequate powers | Mechnical Break down | Less water discharge | Other Reasons | Total (5 to 8) | |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 1 | Bardez | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2 | Bicholim | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 3 | Pernem | 2 | 2 | 0 | 0 | 0 | 0 | 0 | 2 |
| 4 | Ponda | 7 | 7 | 0 | 0 | 0 | 0 | 0 | 7 |
| 5 | Sattari | 16 | 16 | 0 | 0 | 0 | 0 | 0 | 16 |
| 6 | Tiswadi | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | **North Goa District** | **25** | **25** | **0** | **0** | **0** | **0** | **0** | **25** |
| 7 | Canacona | 13 | 7 | 0 | 0 | 5 | 6 | 11 | 18 |
| 8 | Mormugao | 0 | 4 | 0 | 0 | 0 | 0 | 0 | 4 |
| 9 | Quepem | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 10 | Salcete | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 11 | Sanguem | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | **South Goa District** | **13** | **11** | **0** | **0** | **5** | **6** | **11** | **22** |
| | **Goa State** | **38** | **36** | **0** | **0** | **5** | **6** | **11** | **47** |

## 3.9 Number of Deep Tube Wells according to Water Distribution System

(In Nos.)

| Sl. No. | Name of State/District/Taluka | Drip Irrigation | Sprinklers | Others | Open Water Channel | Under Ground Pipe | Surface Pipe | Total (3 to 5) |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| 1 | Bardez | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2 | Bicholim | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 3 | Pernem | 0 | 0 | 0 | 1 | 0 | 1 | 2 |
| 4 | Ponda | 0 | 4 | 0 | 2 | 0 | 1 | 7 |
| 5 | Sattari | 0 | 1 | 0 | 7 | 0 | 8 | 16 |
| 6 | Tiswadi | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | **North Goa District** | **0** | **5** | **0** | **10** | **0** | **10** | **25** |
| 7 | Canacona | 0 | 2 | 6 | 0 | 0 | 10 | 18 |
| 8 | Mormugao | 0 | 0 | 4 | 0 | 0 | 0 | 4 |
| 9 | Quepem | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 10 | Salcete | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 11 | Sanguem | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | **South Goa District** | **0** | **2** | **10** | **0** | **0** | **10** | **22** |
| | **Goa State** | **0** | **7** | **10** | **10** | **0** | **20** | **47** |

## 3.10 Distribution of Deep Tube Wells according to Ownership

(In Nos.)

| Sl. No. | Name of State/District/Taluka | Govt. owned | Coop. Societies | Panchayat | Group of farmers | Others | Individual Farmers | Total (3 to 7) |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| 1 | Bardez | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2 | Bicholim | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 3 | Pernem | 0 | 0 | 0 | 0 | 1 | 1 | 2 |
| 4 | Ponda | 0 | 0 | 0 | 0 | 2 | 5 | 7 |
| 5 | Sattari | 0 | 1 | 0 | 0 | 0 | 15 | 16 |
| 6 | Tiswadi | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | **North Goa District** | **0** | **1** | **0** | **0** | **3** | **21** | **25** |
| 7 | Canacona | 12 | 0 | 0 | 1 | 1 | 4 | 18 |
| 8 | Mormugao | 0 | 0 | 0 | 1 | 0 | 3 | 4 |
| 9 | Quepem | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 10 | Salcete | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 11 | Sanguem | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | **South Goa District** | **12** | **0** | **0** | **2** | **1** | **7** | **22** |
| | **Goa State** | **12** | **1** | **0** | **2** | **4** | **28** | **47** |

## 3.11 Distribution of Deep Tube Wells according the Source of Finance

(In Nos.)

| Sl. No. | Name of State/District/Taluka | Banks Loan | Govt. Fund | Own Saving | Money Lenders | Others | Total (3 to 7) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| 1 | Bardez | 0 | 0 | 0 | 0 | 0 | 0 |
| 2 | Bicholim | 0 | 0 | 0 | 0 | 0 | 0 |
| 3 | Pernem | 1 | 1 | 0 | 0 | 0 | 2 |
| 4 | Ponda | 0 | 4 | 2 | 0 | 1 | 7 |
| 5 | Sattari | 0 | 11 | 0 | 0 | 5 | 16 |
| 6 | Tiswadi | 0 | 0 | 0 | 0 | 0 | 0 |
| | **North Goa District** | **1** | **16** | **2** | **0** | **6** | **25** |
| 7 | Canacona | 1 | 9 | 6 | 0 | 2 | 18 |
| 8 | Mormugao | 0 | 2 | 2 | 0 | 0 | 4 |
| 9 | Quepem | 0 | 0 | 0 | 0 | 0 | 0 |
| 10 | Salcete | 0 | 0 | 0 | 0 | 0 | 0 |
| 11 | Sanguem | 0 | | 0 | 0 | 0 | 0 |
| | **South Goa District** | **1** | **11** | **8** | **0** | **2** | **22** |
| | **Goa State** | **2** | **27** | **10** | **0** | **8** | **47** |

# 4. SURFACE FLOW IRRIGATION SCHEME

## 4.1 Construction of Surface Flow Scheme over the years

(In Nos.)

| Sl. No. | Name of the State/District/Taluka | Construction of Surface Flow over the years | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | Up to 2000-01 | During 2001-02 | During 2002-03 | During 2003-04 | During 2004-05 | During 2005-06 | During 2006-07 | Total (3 to 9) |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 1 | Bardez | 48 | 2 | 0 | 3 | 0 | 1 | 1 | 55 |
| 2 | Bicholim | 46 | 0 | 0 | 0 | 0 | 0 | 0 | 46 |
| 3 | Pernem | 24 | 1 | 3 | 1 | 0 | 0 | 5 | 34 |
| 4 | Ponda | 563 | 35 | 4 | 6 | 2 | 0 | 2 | 612 |
| 5 | Sattari | 76 | 16 | 0 | 2 | 0 | 1 | 0 | 95 |
| 6 | Tiswadi | 6 | 0 | 0 | 0 | 0 | 0 | 0 | 6 |
| | **North Goa District** | **763** | **54** | **7** | **12** | **2** | **2** | **8** | **848** |
| 7 | Canacona | 231 | 5 | 0 | 0 | 0 | 1 | 2 | 239 |
| 8 | Mormugao | 30 | 0 | 0 | 0 | 0 | 0 | 0 | 30 |
| 9 | Quepem | 175 | 7 | 4 | 4 | 1 | 2 | 2 | 195 |
| 10 | Salcete | 65 | 5 | 1 | 3 | 3 | 1 | 3 | 81 |
| 11 | Sanguem | 291 | 44 | 8 | 3 | 0 | 0 | 2 | 348 |
| | **South Goa District** | **792** | **61** | **13** | **10** | **4** | **4** | **9** | **893** |
| | **Goa State** | **1555** | **115** | **20** | **22** | **6** | **6** | **17** | **1741** |

## 4.2 Number of Surface Flow Schemes by Type

(In Nos.)

| Sl. No. | Name of the State/District/Taluka | Type of Surface Flow Schemes | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | Reservoirs | Tanks | Other storage | Permanent Diversion | Temporary Diversion | Water Conservation | Spring Channel | Others | Total (3 to 10) |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
| 1 | Bardez | 15 | 4 | 2 | 23 | 2 | 3 | **6** | **0** | 55 |
| 2 | Bicholim | 4 | 6 | 17 | 2 | 7 | 8 | 1 | 1 | 46 |
| 3 | Pernem | 2 | 9 | 3 | 1 | 11 | 3 | 4 | 1 | 34 |
| 4 | Ponda | 140 | 42 | 5 | 162 | 55 | 63 | 134 | 11 | 612 |
| 5 | Sattari | 11 | 9 | 9 | 41 | 11 | 5 | 8 | 1 | 95 |
| 6 | Tiswadi | 1 | 0 | 1 | 0 | 0 | 0 | 1 | 3 | 6 |
| | **North Goa District** | **173** | **70** | **37** | **229** | **86** | **82** | **154** | **17** | **848** |
| 7 | Canacona | 1 | 0 | 6 | 187 | 27 | 15 | 3 | 0 | 239 |
| 8 | Mormugao | 0 | 5 | 1 | 5 | 7 | 12 | 0 | 0 | 30 |
| 9 | Quepem | 3 | 7 | 5 | 51 | 51 | 11 | 45 | 22 | 195 |
| 10 | Salcete | 4 | 7 | 28 | 11 | 9 | 11 | 11 | 0 | 81 |
| 11 | Sanguem | 1 | 275 | 19 | 33 | 11 | 2 | 7 | 0 | 348 |
| | **South Goa District** | **9** | **294** | **59** | **287** | **105** | **51** | **66** | 22 | **893** |
| | **Goa State** | **182** | **364** | **96** | **516** | **191** | **133** | **220** | 39 | **1741** |

## 4.3 Number of Surface Flow Schemes by Status of Utilisation

(In Nos.)

| Sl. No. | Name of the District/Taluka | Scheme in Use | Surface Flow Schemes not in use | | | |
|---|---|---|---|---|---|---|
| | | | Temporary | Permanent | Total ( 4 to 5) | Grand Total (3+6) |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| 1 | Bardez | 48 | 3 | 4 | 7 | 55 |
| 2 | Bicholim | 45 | 1 | 0 | 1 | 46 |
| 3 | Pernem | 33 | 1 | 0 | 1 | 34 |
| 4 | Ponda | 596 | 15 | 1 | 16 | 612 |
| 5 | Sattari | 54 | 41 | 0 | 41 | 95 |
| 6 | Tiswadi | 6 | 0 | 0 | 0 | 6 |
| | **North Goa District** | **782** | **61** | **5** | **66** | **848** |
| 7 | Canacona | 224 | 5 | 10 | 15 | 239 |
| 8 | Mormugao | 30 | 0 | 0 | 0 | 30 |
| 9 | Quepem | 186 | 3 | 6 | 9 | 195 |
| 10 | Salcete | 78 | 2 | 1 | 3 | 81 |
| 11 | Sanguem | 343 | 5 | 0 | 5 | 348 |
| | **South Goa District** | **861** | **15** | **17** | **32** | **893** |
| | **Goa State** | **1643** | **76** | **22** | **98** | **1741** |

## 4.3 (A) Number of Surface Flow Scheme Temporary not in use by Reasons

(In Nos.)

| Sl. No. | Name of the State/District/Taluka | Surface Flow Schemes Temporary not in use due to | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | Non Availability of Adequate power | Mechanical break Down | Less discharge in the well | Storage not filled up fully | Siltation of Channnel/ Storage | Chanel break down | Others | Total (3 to 9) |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 1 | Bardez | 0 | 0 | 0 | 0 | 0 | 0 | 3 | 3 |
| 2 | Bicholim | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 |
| 3 | Pernem | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
| 4 | Ponda | 0 | 1 | 8 | 1 | 0 | 1 | 4 | 15 |
| 5 | Sattari | 0 | 1 | 0 | 0 | 0 | 5 | 35 | 41 |
| 6 | Tiswadi | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | **North Goa District** | **0** | **2** | **9** | **1** | **0** | **6** | **43** | **61** |
| 7 | Canacona | 0 | 0 | 0 | 0 | 0 | 0 | 5 | **5** |
| 8 | Mormugao | 0 | 0 | 0 | 0 | 0 | 0 | 0 | **0** |
| 9 | Quepem | 0 | 1 | 2 | 0 | 0 | 0 | 1 | **4** |
| 10 | Salcete | 0 | 1 | 0 | 0 | 0 | 0 | 0 | **1** |
| 11 | Sanguem | 3 | 2 | 0 | 0 | 0 | 0 | 0 | **5** |
| | **South Goa District** | **3** | **4** | **2** | **0** | **0** | **0** | **6** | **15** |
| | **Goa State** | **3** | **6** | **11** | **1** | **0** | **6** | **49** | **76** |

## 4.3 (B) Number of Surface Flow Scheme Permanent not in use by Reasons

(In Nos.)

| Sl. No. | Name of the State/District/Taluka | Surface Flow Schemes Temporary not in use due to | | | | | |
|---|---|---|---|---|---|---|---|
| | | Salinity | Dried Up | Destroyed beyond repair | Sinking | Others reasons | Total (3 to 7 ) |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| 1 | Bardez | 0 | 0 | 0 | 0 | 4 | 4 |
| 2 | Bicholim | 0 | 0 | 0 | 0 | 0 | 0 |
| 3 | Pernem | 0 | 0 | 0 | 0 | 0 | 0 |
| 4 | Ponda | 0 | 0 | 1 | 0 | 0 | 1 |
| 5 | Sattari | 0 | 0 | 0 | 0 | 0 | 0 |
| 6 | Tiswadi | 0 | 0 | 0 | 0 | 0 | 0 |
| | **North Goa District** | **0** | **0** | **1** | **0** | **4** | **5** |
| 7 | Canacona | 0 | 1 | 2 | 0 | 7 | 10 |
| 8 | Mormugao | 0 | 0 | 0 | 0 | 0 | 0 |
| 9 | Quepem | 0 | 3 | 0 | 0 | 3 | 6 |
| 10 | Salcete | 0 | 0 | 0 | 0 | 1 | 1 |
| 11 | Sanguem | 0 | 0 | 0 | 0 | 0 | 0 |
| | **South Goa District** | **0** | **4** | **2** | **0** | **11** | 17 |
| | **Goa State** | **0** | **4** | **3** | **0** | **15** | **22** |

## 4.4 Number of Surface Flow Scheme by Owners Holding Size

(In Nos.)

| Sl. No. | Name of the State/District/Taluka | Number of Surface Flow Lift with farmers in Holding Group | | | | | |
|---|---|---|---|---|---|---|---|
| | | Marginal 0-1 | Small 1-2 | Semi Medium 2-4 | Medium 4-10 | Large (Big) >=10 | Total (3 to 7 ) |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| 1 | Bardez | 51 | 4 | 0 | 0 | 0 | 55 |
| 2 | Bicholim | 45 | 1 | 0 | 0 | 0 | 46 |
| 3 | Pernem | 33 | 1 | 0 | 0 | 0 | 34 |
| 4 | Ponda | 575 | 16 | 11 | 7 | 3 | 612 |
| 5 | Sattari | 47 | 25 | 18 | 4 | 1 | 95 |
| 6 | Tiswadi | 6 | 0 | 0 | 0 | 0 | 6 |
| | **North Goa District** | **757** | **47** | **29** | **11** | **4** | **848** |
| 7 | Canacona | 233 | 3 | 3 | 0 | 0 | 239 |
| 8 | Mormugao | 29 | 1 | 0 | 0 | 0 | 30 |
| 9 | Quepem | 176 | 9 | 1 | 6 | 3 | 195 |
| 10 | Salcete | 79 | 1 | 1 | 0 | 0 | 81 |
| 11 | Sanguem | 140 | 85 | 108 | 13 | 2 | 348 |
| | **South Goa District** | **657** | **99** | **113** | **19** | **5** | **893** |
| | **Goa State** | **1414** | **146** | **142** | **30** | **9** | **1741** |

## 4.5 Number of Surface Flow Scheme according to the Social Status

(In Nos.)

| Sl. No. | Name of the State/District/Taluka | Number of Surface Flow Scheme belong to | | | | | Owned by other than individual/ group of farmers | Grand Total (7 + 8) |
|---|---|---|---|---|---|---|---|---|
| | | SC | ST | Others | OBC | Total (3 to 6) | | |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| 1 | Bardez | 0 | 0 | 9 | 3 | 12 | 43 | 55 |
| 2 | Bicholim | 0 | 0 | 5 | 1 | 6 | 40 | 46 |
| 3 | Pernem | 0 | 0 | 1 | 1 | 2 | 32 | 34 |
| 4 | Ponda | 12 | 98 | 72 | 61 | 243 | 369 | 612 |
| 5 | Sattari | 0 | 1 | 61 | 1 | 63 | 32 | 95 |
| 6 | Tiswadi | 0 | 0 | 1 | 0 | 1 | 5 | 6 |
| | **North Goa District** | **12** | **99** | **149** | **67** | **327** | **521** | **848** |
| 7 | Canacona | 0 | 14 | 5 | 1 | 20 | 219 | 239 |
| 8 | Mormugao | 0 | 11 | 7 | 2 | 20 | 10 | 30 |
| 9 | Quepem | 1 | 40 | 41 | 3 | 85 | 110 | 195 |
| 10 | Salcete | 9 | 14 | 30 | 2 | 55 | 26 | 81 |
| 11 | Sanguem | 27 | 173 | 90 | 12 | 302 | 46 | 348 |
| | **South Goa District** | **37** | **252** | **173** | **20** | **482** | **411** | **893** |
| | **Goa State** | **49** | **351** | **322** | **87** | **722** | **932** | **1741** |

## 4.6 Culturable Command Area and gross Irrigation Potential Created & Utilized by Surface Flow Scheme

(In ha.)

| Sl. No. | Name of the State/District/Taluka | No of Surface Flow Scheme | | Culturable Command area | Irrigation potential created by Surface Flow Scheme | | | | | Irrigation potential utilised by Surface Flow Scheme | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | In use | Not in use | | Kharif | Rabi | Perinnial | Others | Total (6 to 9) | Kharif | Rabi | Perinnial | Others | Total (11 to 14) |
| **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** | **9** | **10** | **11** | **12** | **13** | **14** | **15** |
| 1 | Bardez | 48 | 7 | 169 | 9 | 37 | 22 | 108 | 176 | 8 | 36 | 22 | 108 | 174 |
| 2 | Bicholim | 45 | 1 | 414 | 0 | 330 | 85 | 0 | 415 | 0 | 328 | 85 | 0 | 413 |
| 3 | Pernem | 33 | 1 | 131 | 16 | 94 | 28 | 0 | 138 | 11 | 94 | 28 | 0 | 133 |
| 4 | Ponda | 596 | 16 | 1189 | 71 | 483 | 649 | 0 | 1203 | 71 | 461 | 623 | 0 | 1155 |
| 5 | Sattari | 54 | 41 | 221 | 4 | 89 | 126 | 2 | 221 | 4 | 68 | 98 | 1 | 171 |
| 6 | Tiswadi | 6 | 0 | 75 | 73 | 44 | 0 | 0 | 117 | 38 | 28 | 0 | 0 | 66 |
| | **North Goa District** | **782** | **66** | **2199** | **173** | **1077** | **910** | **110** | **2270** | **132** | **1015** | **856** | **109** | **2112** |
| 7 | Canacona | 224 | 15 | 438 | 361 | 150 | 13 | 1 | 525 | 333 | 117 | 12 | 1 | 463 |
| 8 | Mormugao | 30 | 0 | 21 | 1 | 20 | 0 | 0 | 21 | 1 | 20 | 0 | 0 | 21 |
| 9 | Quepem | 186 | 9 | 381 | 0 | 295 | 55 | 31 | 381 | 0 | 272 | 54 | 31 | 357 |
| 10 | Salcete | 78 | 3 | 75 | 45 | 41 | 0 | 0 | 86 | 33 | 40 | 0 | 0 | 73 |
| 11 | Sanguem | 343 | 5 | 754 | 0 | 431 | 305 | 22 | 758 | 0 | 428 | 305 | 22 | 755 |
| | **South Goa District** | **861** | **32** | **1669** | **407** | **937** | **373** | **54** | **1771** | **367** | **877** | **371** | **54** | **1669** |
| | **Goa State** | **1643** | **98** | **3868** | **580** | **2014** | **1283** | **164** | **4041** | **499** | **1892** | **1227** | **163** | **3781** |

## 4.7 Area Irrigated from other sources but within the command area of Surface Flow Scheme

(In ha.)

| Sl. No. | Name of the State/District/Taluka | No of Surface Flow Scheme | Culturable Command area | Net Area Irrigated from other sources but within the command area of Surface Flow Scheme | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | Kharif | Rabi | Perinnial | Others | Total (5 to 8) |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| 1 | Bardez | 55 | 169 | 0 | 0 | 0 | 0 | 0 |
| 2 | Bicholim | 46 | 414 | 0 | 0 | 0 | 0 | 0 |
| 3 | Pernem | 34 | 131 | 0 | 0 | 0 | 0 | 0 |
| 4 | Ponda | 612 | 1189 | 0 | 0 | 0 | 0 | 0 |
| 5 | Sattari | 95 | 221 | 0 | 0 | 0 | 0 | 0 |
| 6 | Tiswadi | 6 | 75 | 0 | 0 | 0 | 0 | 0 |
| | **North Goa District** | **848** | **2199** | **0** | **0** | **0** | **0** | **0** |
| 7 | Canacona | 239 | 438 | 1 | 1 | 0 | 0 | 2 |
| 8 | Mormugao | 30 | 21 | 0 | 0 | 0 | 0 | 0 |
| 9 | Quepem | 195 | 381 | 0 | 147 | 2 | 0 | 149 |
| 10 | Salcete | 81 | 75 | 16 | 17 | 0 | 0 | 33 |
| 11 | Sanguem | 348 | 754 | 0 | 0 | 0 | 0 | 0 |
| | **South Goa District** | **893** | **1669** | **17** | **165** | **2** | **0** | **184** |
| | **Goa State** | **1741** | **3868** | **17** | **165** | **2** | **0** | **184** |

## 4.8 No of Surface Flow Scheme according to the Constraints

(In Nos.)

| Sl. No. | Name of the State/District/ Taluka | No of Surface Flow Scheme in use | No of Surface Flow Scheme with no constraints | Constraints in utilisation of Irrigation potential created | | | | | | | Total (5 to 11) | Grand Total (4+12) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Inade-quate power supply | Mechan-ical Break down | Less water discha-rge | Storage not filled up fully | Siltation of Canal/ Storage | Channel Break Down | Other reasons | | |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 |
| 1 | Bardez | 48 | 54 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 55 |
| 2 | Bicholim | 45 | 46 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 46 |
| 3 | Pernem | 33 | 34 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 34 |
| 4 | Ponda | 596 | 592 | 0 | 1 | 3 | 0 | 0 | 0 | 16 | 20 | 612 |
| 5 | Sattari | 54 | 93 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 2 | 95 |
| 6 | Tiswadi | 6 | 5 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 6 |
| | **North Goa District** | **782** | **824** | **0** | **2** | **3** | **0** | **1** | **0** | **18** | **24** | **848** |
| 7 | Canacona | 224 | 219 | 0 | 4 | 0 | 0 | 0 | 2 | 14 | 20 | 239 |
| 8 | Mormugao | 30 | 30 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 30 |
| 9 | Quepem | 186 | 186 | 0 | 1 | 0 | 0 | 0 | 0 | 8 | 9 | 195 |
| 10 | Salcete | 78 | 27 | 0 | 1 | 1 | 0 | 0 | 0 | 52 | 54 | 81 |
| 11 | Sanguem | 343 | 346 | 0 | 2 | 0 | 0 | 0 | 0 | 0 | 2 | 348 |
| | **South Goa District** | **861** | **808** | **0** | **8** | **1** | **0** | **0** | **2** | **74** | **85** | **893** |
| | **Goa State** | **1643** | **1632** | **0** | **10** | **4** | **0** | **1** | **2** | **92** | **109** | **1741** |

## 4.9 Number of Surface Flow Schemes according to Water Distribution System

(In Nos.)

| Sl. No. | Name of State/District/Taluka | Sprinklers | Drip Irrigation | Open Water Channel | Surface pipe | Under Gr. Channel | Others | Total (3 to 7) |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| 1 | Bardez | 0 | 0 | 36 | 5 | 0 | 14 | **55** |
| 2 | Bicholim | 0 | 0 | 34 | 1 | 0 | 11 | **46** |
| 3 | Pernem | 0 | 0 | 32 | 1 | 0 | 1 | **34** |
| 4 | Ponda | 2 | 0 | 442 | 1 | 0 | 167 | **612** |
| 5 | Sattari | 1 | 0 | 51 | 0 | 0 | 43 | **95** |
| 6 | Tiswadi | 0 | 0 | 3 | 0 | 2 | 1 | **6** |
| | **North Goa District** | **3** | **0** | **598** | **8** | **2** | **237** | **848** |
| 7 | Canacona | 0 | 0 | 217 | 2 | 0 | 20 | **239** |
| 8 | Mormugao | 0 | 0 | 20 | 2 | 0 | 8 | **30** |
| 9 | Quepem | 1 | 1 | 175 | 1 | 0 | 17 | **195** |
| 10 | Salcete | 0 | 0 | 20 | 2 | 8 | 51 | **81** |
| 11 | Sanguem | 0 | 0 | 340 | 2 | 0 | 6 | **348** |
| | **South Goa District** | **1** | **1** | **772** | **9** | **8** | **102** | **893** |
| | **Goa State** | **4** | **1** | **1370** | **17** | **10** | **339** | **1741** |

## 4.10 Distribution of Surface Flow Schemes according to Ownership

(In Nos.)

| Sl. No. | Name of State/District/Taluka | Govt. owned | Coop. Societies | Panchayat | Group of farmers | Individual farmers | Others | Total (3 to 8) |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| 1 | Bardez | 31 | 0 | 0 | 9 | 12 | 3 | 55 |
| 2 | Bicholim | 0 | 0 | 0 | 40 | 6 | 0 | 46 |
| 3 | Pernem | 6 | 0 | 0 | 25 | 2 | 1 | 34 |
| 4 | Ponda | 11 | 0 | 1 | 354 | 243 | 3 | 612 |
| 5 | Sattari | 1 | 1 | 0 | 21 | 63 | 9 | 95 |
| 6 | Tiswadi | 5 | 0 | 0 | 0 | 1 | 0 | 6 |
| | **North Goa District** | **54** | **1** | **1** | **449** | **327** | **16** | **848** |
| 7 | Canacona | 24 | 0 | 0 | 185 | 20 | 10 | 239 |
| 8 | Mormugao | 0 | 0 | 0 | 10 | 20 | 0 | 30 |
| 9 | Quepem | 3 | 0 | 0 | 95 | 85 | 12 | 195 |
| 10 | Salcete | 0 | 0 | 0 | 22 | 55 | 4 | 81 |
| 11 | Sanguem | 2 | 5 | 0 | 39 | 302 | 0 | 348 |
| | **South Goa District** | **29** | **5** | **0** | **351** | **482** | **26** | **893** |
| | **Goa State** | **83** | **6** | **1** | **800** | **809** | **42** | **1741** |

## 4.11 Distribution of Dug wells according the Source of Finance

(In Nos.)

| Sl. No. | Name of State/District/Taluka | Banks Loan | Govt. Fund | Own Saving | Money Lenders | Others | Total (3 to 7) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| 1 | Bardez | 7 | 22 | 13 | 2 | 11 | 55 |
| 2 | Bicholim | 1 | 4 | 15 | 0 | 26 | 46 |
| 3 | Pernem | 0 | 17 | 4 | 0 | 13 | 34 |
| 4 | Ponda | 11 | 29 | 471 | 7 | 94 | 612 |
| 5 | Sattari | 5 | 7 | 68 | 0 | 15 | 95 |
| 6 | Tiswadi | 0 | 1 | 1 | 0 | 4 | 6 |
| | **North Goa District** | **24** | **80** | **572** | **9** | **163** | **848** |
| 7 | Canacona | 5 | 19 | 166 | 0 | 49 | 239 |
| 8 | Mormugao | 0 | 5 | 25 | 0 | 0 | 30 |
| 9 | Quepem | 8 | 4 | 173 | 3 | 7 | 195 |
| 10 | Salcete | 0 | 5 | 74 | 0 | 2 | 81 |
| 11 | Sanguem | 207 | 88 | 53 | 0 | 0 | 348 |
| | **South Goa District** | **220** | **121** | **491** | **3** | **58** | **893** |
| | **Goa State** | **244** | **201** | **1063** | **12** | **221** | **1741** |

# 5. SURFACE LIFT SCHEMES

## 5.1 Construction of Surface Lift Schemes over the years

(In Nos.)

| Sl. No. | Name of the State/District/Taluka | Construction of Deep Tube Wells over the years | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | Up to 2000-01 | During 2001-02 | During 2002-03 | During 2003-04 | During 2004-05 | During 2005-06 | During 2006-07 | Total (3 to 10) |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 11 |
| 1 | Bardez | 10 | 0 | 0 | 0 | 0 | 0 | 0 | 10 |
| 2 | Bicholim | 65 | 5 | 3 | 3 | 0 | 14 | 0 | 90 |
| 3 | Pernem | 48 | 4 | 3 | 3 | 0 | 2 | 7 | 67 |
| 4 | Ponda | 49 | 3 | 15 | 1 | 0 | 1 | 10 | 79 |
| 5 | Sattari | 83 | 3 | 1 | 1 | 4 | 1 | 1 | 94 |
| 6 | Tiswadi | 5 | 0 | 1 | 0 | 0 | 0 | 0 | 6 |
| | **North Goa District** | **260** | **15** | **23** | **8** | **4** | **18** | **18** | **346** |
| 7 | Canacona | 25 | 5 | 3 | 1 | 1 | 1 | 3 | 39 |
| 8 | Mormugao | 2 | 0 | 0 | 0 | 0 | 0 | 6 | 8 |
| 9 | Quepem | 58 | 2 | 4 | 4 | 4 | 4 | 10 | 86 |
| 10 | Salcete | 8 | 0 | 0 | 0 | 0 | 1 | 4 | 13 |
| 11 | Sanguem | 408 | 6 | 2 | 0 | 1 | 0 | 1 | 418 |
| | **South Goa District** | **501** | **13** | **9** | **5** | **6** | **6** | **24** | **564** |
| | **Goa State** | **761** | **28** | **32** | **13** | **10** | **24** | **42** | **910** |

## 5.2 Number of Surface Flow Schemes by Type

(In Nos.)

| Sl. No. | Name of the State/District/Taluka | Type of Surface Lift Schemes | | | | | |
|---|---|---|---|---|---|---|---|
| | | On river | On Stream | On Drain/ Canal | On tank/Pond | Others | Total (3 to 7) |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| 1 | Bardez | 1 | 1 | 0 | 7 | 1 | 10 |
| 2 | Bicholim | 57 | 0 | 5 | 10 | 18 | 90 |
| 3 | Pernem | 46 | 6 | 0 | 11 | 4 | 67 |
| 4 | Ponda | 24 | 19 | 0 | 29 | 7 | 79 |
| 5 | Sattari | 65 | 8 | 1 | 20 | 0 | 94 |
| 6 | Tiswadi | 0 | 0 | 0 | 0 | 6 | 6 |
| | **North Goa District** | **193** | **34** | **6** | **77** | **36** | **346** |
| 7 | Canacona | 22 | 5 | 0 | 0 | 12 | 39 |
| 8 | Mormugao | 1 | 1 | 0 | 1 | 5 | 8 |
| 9 | Quepem | 59 | 7 | 13 | 0 | 7 | 86 |
| 10 | Salcete | 1 | 2 | 4 | 2 | 4 | 13 |
| 11 | Sanguem | 395 | 5 | 0 | 18 | 0 | 418 |
| | **South Goa District** | **478** | **20** | **17** | **21** | **28** | **564** |
| | **Goa State** | **671** | **54** | **23** | **98** | **64** | **910** |

## 5.3 Number of Surface Lift Schemes by Status of Utilisation

(In Nos.)

| Sl. No. | Name of the State/District/Taluka | Scheme in use | Surface Lift Schemes not in use | | | |
|---|---|---|---|---|---|---|
| | | | Temporary | Permanent | Total (4 to 5) | Grand Total (3+6) |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| 1 | Bardez | 8 | 1 | 1 | 2 | 10 |
| 2 | Bicholim | 89 | 1 | 0 | 1 | 90 |
| 3 | Pernem | 66 | 1 | 0 | 1 | 67 |
| 4 | Ponda | 77 | 0 | 2 | 2 | 79 |
| 5 | Sattari | 93 | 1 | 0 | 1 | 94 |
| 6 | Tiswadi | 5 | 0 | 1 | 1 | 6 |
| | **North Goa District** | **338** | **4** | **4** | **8** | **346** |
| 7 | Canacona | 32 | 7 | 0 | 7 | 39 |
| 8 | Mormugao | 8 | 0 | 0 | 0 | 8 |
| 9 | Quepem | 85 | 0 | 1 | 1 | 86 |
| 10 | Salcete | 13 | 0 | 0 | 0 | 13 |
| 11 | Sanguem | 416 | 1 | 1 | 2 | 418 |
| | **South Goa District** | **554** | **8** | **2** | **10** | **564** |
| | **Goa State** | **892** | **12** | **6** | **18** | **910** |

## 5.3 (A) Number of Surface Lift Scheme Temporary not in use by Reasons

(In Nos.)

| Sl. No. | Name of the State/District/Taluka | Surface Lift Schemes Temporary not in use due to | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | Non Availability of Adequate power | Mechanical break Down | Less discharge in the well | Storage not filled up fully | Siltation of Channel/ Storage | Channel break down | Others | Total (3 to 9) |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 1 | Bardez | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
| 2 | Bicholim | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
| 3 | Pernem | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
| 4 | Ponda | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 5 | Sattari | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 |
| 6 | Tiswadi | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | **North Goa District** | **0** | **1** | **0** | **0** | **0** | **0** | **3** | **4** |
| 7 | Canacona | 0 | 1 | 0 | 0 | 0 | 0 | 6 | **7** |
| 8 | Mormugao | 0 | 0 | 0 | 0 | 0 | 0 | 0 | **0** |
| 9 | Quepem | 0 | 0 | 0 | 0 | 0 | 0 | 0 | **0** |
| 10 | Salcete | 0 | 0 | 0 | 0 | 0 | 0 | 0 | **0** |
| 11 | Sanguem | 0 | 1 | 0 | 0 | 0 | 0 | 0 | **1** |
| | **South Goa District** | **0** | **2** | **0** | **0** | **0** | **0** | **6** | **8** |
| | **Goa State** | **0** | **3** | **0** | **0** | **0** | **0** | **9** | **12** |

## 5.3 (B) Number of Surface Lift Scheme Permanent not in use by Reasons

(In Nos.)

| Sl. No. | Name of the State/District/Taluka | Surface Lift Schemes Temporary not in use due to | | | | | |
|---|---|---|---|---|---|---|---|
| | | Salinity | Dried Up | Destroyed beyond repair | Sinking | Others reasons | Total (3 to 7 ) |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| 1 | Bardez | 0 | 0 | 0 | 0 | 1 | 1 |
| 2 | Bicholim | 0 | 0 | 0 | 0 | 0 | 0 |
| 3 | Pernem | 0 | 0 | 0 | 0 | 0 | 0 |
| 4 | Ponda | 0 | 0 | 0 | 0 | 2 | 2 |
| 5 | Sattari | 0 | 0 | 0 | 0 | 0 | 0 |
| 6 | Tiswadi | 0 | 0 | 0 | 0 | 1 | 1 |
| | **North Goa District** | **0** | **0** | **0** | **0** | **4** | **4** |
| 7 | Canacona | 0 | 0 | 0 | 0 | 0 | 0 |
| 8 | Mormugao | 0 | 0 | 0 | 0 | 0 | 0 |
| 9 | Quepem | 0 | 0 | 0 | 0 | 1 | 1 |
| 10 | Salcete | 0 | 0 | 0 | 0 | 0 | 0 |
| 11 | Sanguem | 0 | 0 | 0 | 0 | 1 | 1 |
| | **South Goa District** | **0** | **0** | **0** | **0** | **2** | 2 |
| | **Goa State** | **0** | **0** | **0** | **0** | **6** | **6** |

## 5.4 Number of Surface Flow Scheme by Owners Holding Size

(In Nos.)

| Sl. No. | Name of the State/District/Taluka | Number of Surface Flow Lift with farmers in Holding Group | | | | | |
|---|---|---|---|---|---|---|---|
| | | Marginal 0-1 | Small 1-2 | Semi Medium 2-4 | Medium 4-10 | Large (Big) >=10 | Total (3 to 7 ) |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| 1 | Bardez | 10 | 0 | 0 | 0 | 0 | 10 |
| 2 | Bicholim | 53 | 14 | 9 | 12 | 2 | 90 |
| 3 | Pernem | 56 | 2 | 3 | 6 | 0 | 67 |
| 4 | Ponda | 53 | 9 | 6 | 11 | 0 | 79 |
| 5 | Sattari | 51 | 11 | 17 | 11 | 4 | 94 |
| 6 | Tiswadi | 6 | 0 | 0 | 0 | 0 | 6 |
| | **North Goa District** | **229** | **36** | **35** | **40** | **6** | **346** |
| 7 | Canacona | 27 | 6 | 4 | 2 | 0 | 39 |
| 8 | Mormugao | 6 | 2 | 0 | 0 | 0 | 8 |
| 9 | Quepem | 27 | 27 | 9 | 10 | 13 | 86 |
| 10 | Salcete | 10 | 1 | 2 | 0 | 0 | 13 |
| 11 | Sanguem | 15 | 115 | 187 | 90 | 11 | 418 |
| | **South Goa District** | **85** | **151** | **202** | **102** | **24** | **564** |
| | **Goa State** | **314** | **187** | **237** | **142** | **30** | **910** |

## 5.5 Number of Surface Lift Schemes according to the Social Status

| Sl. No. | Name of the State/District/Taluka | Number of Surface Lift Schemes belong to | | | | | Owned by other than individual/ group of farmers | Grand Total (7 + 8) |
|---|---|---|---|---|---|---|---|---|
| | | SC | ST | Others | OBC | Total (3 to 6) | | |
| **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** | **8** |
| 1 | Bardez | 1 | 0 | 1 | 0 | 2 | 8 | 10 |
| 2 | Bicholim | 1 | 0 | 78 | 2 | 81 | 9 | 90 |
| 3 | Pernem | 0 | 0 | 42 | 6 | 48 | 19 | 67 |
| 4 | Ponda | 0 | 10 | 44 | 9 | 63 | 16 | 79 |
| 5 | Sattari | 0 | 6 | 48 | 0 | 54 | 40 | 94 |
| 6 | Tiswadi | 1 | 2 | 1 | 2 | 6 | 0 | 6 |
| | **North Goa District** | **3** | **18** | **214** | **19** | **254** | **92** | **346** |
| 7 | Canacona | 0 | 15 | 16 | 0 | 31 | 8 | 39 |
| 8 | Mormugao | 0 | 0 | 8 | 0 | 8 | 0 | 8 |
| 9 | Quepem | 0 | 16 | 54 | 2 | 72 | 14 | 86 |
| 10 | Salcete | 0 | 1 | 7 | 1 | 9 | 4 | 13 |
| 11 | Sanguem | 5 | 170 | 239 | 2 | 416 | 2 | 418 |
| | **South Goa District** | **5** | **202** | **324** | **5** | **536** | **28** | **564** |
| | **Goa State** | **8** | **220** | **538** | **24** | **790** | **120** | **910** |

## 5.6 Number of Surface Lift Schemes by Source of Energy

(In Nos.)

| Sl. No. | Name of the State/District/Taluka | No of Deep Tube Wells with Lifting Devices | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | Electric Pump | Diesel | Wind mills | Solar Pumps | Manual/ Animal | Others | Total (3 to 8) |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| 1 | Bardez | 4 | 0 | 0 | 0 | 4 | 2 | 10 |
| 2 | Bicholim | 63 | 9 | 0 | 0 | 0 | 18 | 90 |
| 3 | Pernem | 27 | 39 | 0 | 0 | 0 | 1 | 67 |
| 4 | Ponda | 66 | 0 | 0 | 0 | 11 | 2 | 79 |
| 5 | Sattari | 80 | 12 | 0 | 0 | 1 | 1 | 94 |
| 6 | Tiswadi | 0 | 0 | 0 | 0 | 5 | 1 | 6 |
| | **North Goa District** | **240** | **60** | **0** | **0** | **21** | **25** | **346** |
| 7 | Canacona | 25 | 1 | 1 | 0 | 5 | 7 | 39 |
| 8 | Mormugao | 2 | 0 | 0 | 3 | 1 | 2 | 8 |
| 9 | Quepem | 78 | 6 | 0 | 0 | 1 | 1 | 86 |
| 10 | Salcete | 2 | 3 | 0 | 0 | 7 | 1 | 13 |
| 11 | Sanguem | 407 | 9 | 0 | 0 | 0 | 2 | 418 |
| | **South Goa District** | **514** | **19** | **1** | **3** | **14** | **13** | **564** |
| | **Goa State** | **754** | **79** | **1** | **3** | **35** | **38** | **910** |

## 5.7 Culturable command area and gross irrigation potential created & utilised by Surface Lift Schemes

(In ha.)

| Sl. No. | Name of the State/District/Taluka | No of Surface Lift Scheme | | Culturable Command area | Irrigation potential created by Surface Lift Scheme | | | | | Irrigation potential utilised by Surface Lift Scheme | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | In Use | Not in use | | Kharif | Rabi | Perinnial | Others | Total (6 to 9) | Kharif | Rabi | Perinnial | Others | Total (11 to 14) |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
| 1 | Bardez | 8 | 2 | 4 | 1 | 4 | 0 | 0 | 5 | 1 | 4 | 0 | 0 | 5 |
| 2 | Bicholim | 89 | 1 | 291 | 0 | 41 | 249 | 1 | 291 | 0 | 42 | 248 | 1 | 291 |
| 3 | Pernem | 66 | 1 | 151 | 68 | 95 | 14 | 1 | 178 | 67 | 92 | 14 | 1 | 174 |
| 4 | Ponda | 77 | 2 | 215 | 9 | 17 | 144 | 58 | 228 | 9 | 17 | 143 | 58 | 227 |
| 5 | Sattari | 93 | 1 | 602 | 0 | 57 | 545 | 7 | 609 | 0 | 49 | 525 | 3 | 577 |
| 6 | Tiswadi | 5 | 1 | 2 | 2 | 2 | 0 | 0 | 4 | 2 | 2 | 0 | 0 | 4 |
| | **North Goa District** | **338** | **8** | **1265** | **80** | **216** | **952** | **67** | **1315** | **79** | **206** | **930** | **63** | **1278** |
| 7 | Canacona | 32 | 7 | 82 | 23 | 68 | 2 | 0 | 93 | 16 | 38 | 1 | 0 | 55 |
| 8 | Mormugao | 8 | 0 | 4 | 0 | 4 | 0 | 0 | 4 | 0 | 4 | 0 | 0 | 4 |
| 9 | Quepem | 85 | 1 | 487 | 0 | 55 | 432 | 0 | 487 | 0 | 55 | 431 | 0 | 486 |
| 10 | Salcete | 13 | 0 | 6 | 2 | 3 | 0 | 0 | 5 | 2 | 3 | 0 | 0 | 5 |
| 11 | Sanguem | 416 | 2 | 1289 | 17 | 54 | 1213 | 4 | 1288 | 17 | 53 | 1211 | 4 | 1285 |
| | **South Goa District** | **554** | **10** | **1868** | **42** | **184** | **1647** | **4** | **1877** | **35** | **153** | **1643** | **4** | **1835** |
| | **Goa State** | **892** | **18** | **3133** | **122** | **400** | **2599** | **71** | **3192** | **114** | **359** | **2573** | **67** | **3113** |

## 5.8 Area Irrigated from other sources but within the command area of Surface Lift Schemes

(In ha.)

| Sl. No. | Name of the State/District/Taluka | No of Surface Lift Schemes | Culturable Command area | Net Area Irrigated from other sources but within the command area of Surface Lift Schemes | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | Kharif | Rabi | Perinnial | Others | Total (5 to 8) |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| 1 | Bardez | 10 | 4 | 0 | 0 | 0 | 0 | 0 |
| 2 | Bicholim | 90 | 291 | 0 | 1 | 16 | 0 | 17 |
| 3 | Pernem | 67 | 151 | 0 | 0 | 0 | 0 | 0 |
| 4 | Ponda | 79 | 215 | 0 | 0 | 0 | 0 | 0 |
| 5 | Sattari | 94 | 602 | 0 | 0 | 0 | 0 | 0 |
| 6 | Tiswadi | 6 | 2 | 0 | 0 | 0 | 0 | 0 |
| | **North Goa District** | **346** | **1265** | **0** | **1** | **16** | **0** | **17** |
| 7 | Canacona | 39 | 82 | 0 | 2 | 0 | 0 | 2 |
| 8 | Mormugao | 8 | 4 | 0 | 0 | 0 | 0 | 0 |
| 9 | Quepem | 86 | 487 | 0 | 35 | 7 | 0 | 42 |
| 10 | Salcete | 13 | 6 | 0 | 0 | 0 | 0 | 0 |
| 11 | Sanguem | 418 | 1289 | 0 | 0 | 0 | 0 | 0 |
| | **South Goa District** | **564** | **1868** | **0** | **37** | **7** | **0** | **44** |
| | **Goa State** | **910** | **3133** | **0** | **38** | **23** | **0** | **61** |

## 5.9 No of Surface Lift Schemes according to the Constraints

(In Nos.)

| Sl. No. | Name of the State/District/Taluka | No of Surface Lift Schemes in use | No of Surface Lift Schemes with no constraints | Constraints in utilisation of Irrigation potential created | | | | | | | | Grand Total (4 + 12) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Inadequate powers | Mechanical Breakdown | Less water discharge | Storage not filled up fully | Siltation of Canal/ Storage | Channel break-down | Other reasons | Total (5 to 11) | |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 |
| 1 | Bardez | 8 | 9 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 10 |
| 2 | Bicholim | 89 | 90 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 90 |
| 3 | Pernem | 66 | 66 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 67 |
| 4 | Ponda | 77 | 79 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 79 |
| 5 | Sattari | 93 | 88 | 0 | 1 | 0 | 0 | 0 | 0 | 5 | 6 | 94 |
| 6 | Tiswadi | 5 | 5 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 6 |
| | **North Goa District** | **338** | **337** | **0** | **1** | **0** | **0** | **0** | **0** | **8** | **9** | **346** |
| 7 | Canacona | 32 | 29 | 0 | 1 | 0 | 0 | 0 | 0 | 9 | 10 | 39 |
| 8 | Mormugao | 8 | 8 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 8 |
| 9 | Quepem | 85 | 85 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 86 |
| 10 | Salcete | 13 | 10 | 0 | 0 | 0 | 0 | 0 | 0 | 3 | 3 | 13 |
| 11 | Sanguem | 416 | 417 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 418 |
| | **South Goa District** | **554** | **549** | **0** | **3** | **0** | **0** | **0** | **0** | **12** | **15** | **564** |
| | **Goa State** | **892** | **886** | **0** | **4** | **0** | **0** | **0** | **0** | **20** | **24** | **910** |

## 5.10 Number of Surface Flow Schemes according to Water Distribution System

(In Nos.)

| Sl. No. | Name of State/District/Taluka | Open Channel | Under Gr. Channel | Surface Pipe Channel | Drip Irrigation | Sprinklers | Others | Total (3 to 8) |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| 1 | Bardez | 5 | 0 | 2 | 0 | 0 | 3 | 10 |
| 2 | Bicholim | 25 | 1 | 23 | 3 | 32 | 6 | 90 |
| 3 | Pernem | 60 | 0 | 6 | 0 | 0 | 1 | 67 |
| 4 | Ponda | 39 | 0 | 4 | 1 | 31 | 4 | 79 |
| 5 | Sattari | 52 | 0 | 4 | 5 | 32 | 1 | 94 |
| 6 | Tiswadi | 2 | 0 | 0 | 0 | 0 | 4 | 6 |
| | **North Goa District** | **183** | **1** | **39** | **9** | **95** | **19** | **346** |
| 7 | Canacona | 11 | 0 | 15 | 0 | 2 | 11 | 39 |
| 8 | Mormugao | 1 | 0 | 1 | 0 | 0 | 6 | 8 |
| 9 | Quepem | 65 | 0 | 5 | 2 | 9 | 5 | 86 |
| 10 | Salcete | 8 | 0 | 1 | 0 | 1 | 3 | 13 |
| 11 | Sanguem | 396 | 1 | 9 | 0 | 10 | 2 | 418 |
| | **South Goa District** | **481** | **1** | **31** | **2** | **22** | **27** | **564** |
| | **Goa State** | **664** | **2** | **70** | **11** | **117** | **46** | **910** |

## 5.11 Distribution of Surface Flow Schemes according to Ownership

(In Nos.)

| Sl. No. | Name of State/District/Taluka | Govt. owned | Coop. Societies | Panchayat | Group of farmers | Individual farmers | Others | Total (3 to 8) |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| 1 | Bardez | 2 | 0 | 0 | 6 | 2 | 0 | 10 |
| 2 | Bicholim | 1 | 0 | 0 | 8 | 81 | 0 | 90 |
| 3 | Pernem | 10 | 0 | 2 | 7 | 48 | 0 | 67 |
| 4 | Ponda | 4 | 0 | 0 | 12 | 63 | 0 | 79 |
| 5 | Sattari | 15 | 0 | 0 | 8 | 54 | 17 | 94 |
| 6 | Tiswadi | 0 | 0 | 0 | 0 | 6 | 0 | 6 |
| | **North Goa District** | **32** | **0** | **2** | **41** | **254** | **17** | **346** |
| 7 | Canacona | 1 | 0 | 0 | 3 | 31 | 4 | 39 |
| 8 | Mormugao | 0 | 0 | 0 | 0 | 8 | 0 | 8 |
| 9 | Quepem | 1 | 7 | 0 | 6 | 72 | 0 | 86 |
| 10 | Salcete | 0 | 0 | 0 | 3 | 9 | 1 | 13 |
| 11 | Sanguem | 0 | 1 | 0 | 1 | 416 | 0 | 418 |
| | **South Goa District** | **2** | **8** | **0** | **13** | **536** | **5** | **564** |
| | **Goa State** | **34** | **8** | **2** | **54** | **790** | **22** | **910** |

## 5.12 Distribution of Dug wells according the Source of Finance

(In Nos.)

| Sl. No. | Name of State/District/Taluka | Bank Loan | Govt. Fund | Own Saving | Money Lenders | Others | Total (3 to 6) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| 1 | Bardez | 0 | 3 | 1 | 0 | 6 | 10 |
| 2 | Bicholim | 23 | 7 | 57 | 0 | 3 | 90 |
| 3 | Pernem | 2 | 21 | 44 | 0 | 0 | 67 |
| 4 | Ponda | 6 | 13 | 59 | 0 | 1 | 79 |
| 5 | Sattari | 17 | 22 | 42 | 1 | 12 | 94 |
| 6 | Tiswadi | 0 | 0 | 6 | 0 | 0 | 6 |
| | **North Goa District** | **48** | **66** | **209** | **1** | **22** | **346** |
| 7 | Canacona | 10 | 1 | 25 | 0 | 3 | 39 |
| 8 | Mormugao | 1 | 1 | 6 | 0 | 0 | 8 |
| 9 | Quepem | 5 | 11 | 63 | 0 | 7 | 86 |
| 10 | Salcete | 3 | 1 | 9 | 0 | 0 | 13 |
| 11 | Sanguem | 377 | 20 | 21 | 0 | 0 | 418 |
| | **South Goa District** | **396** | **34** | **124** | **0** | **10** | **564** |
| | **Goa State** | **444** | **100** | **333** | **1** | **32** | **910** |

## FOURTH CENSUS OF MINOR IRRIGATION SCHEMES
## REFERENCE YEAR 2006-07

### SCHEDULE 1 : GROUND WATER SCHEMES

**I. IDENTIFICATION: (USE 2001 POPULATION CENSUS CODE ONLY)**

**State:** _____ **Code :** [ ][ ] **District:** _____ **Code:** [ ][ ]

**Block:** _____ **Code:** [ ][ ][ ] **Village:** _____ **Code:** [ ][ ][ ]

**Date of Enumeration:** [ ][ ] - [ ][ ] - [ ][ ]

**II. SPECIFIC INFORMATION:**

**1. Serial Number of scheme :** [ ][ ][ ]

**2. Type of Scheme** **Code:** [ ]

Dug Well – 1, Shallow Tube well - 2, Deep Tube well - 3

**3. Owner of the Scheme ( Name in case of individual farmer)**

Name _____ **Code:** [ ]

Govt. owned - 1, Co-operative owned - 2, Panchayat owned - 3, Owned by Group of farmers - 4,
Owned by individual farmers - 5, Others – 6

**4. Khasra number /Plot No./Survey No. in which the scheme is located** _____

**5(a). Total Holding of owner (in case of individual owner only)** [ ][ ][ ].[ ][ ] **Ha.**

**(b) Social Status of Owner ( in case of individual owner only)** **Code:** [ ]

Scheduled caste -1 , Scheduled tribe – 2 , OBC- 3, Others 4

**6. Year of Commissioning of the Scheme** **Code :** [ ]

On or before 2000-2001 - 1, during 2001-02 - 2, during 2002-03 - 3, during 2003-04 - 4
during 2004-05 - 5 , during 2005-06 - 6 , during 2006-07 - 7

**7. Nature of scheme** **Code :** [ ]

**Dugwell : Pucca -1, Kuchha -2, Dug-cum-bore well - 3, Others - 4**
**Shallow Tubewell : Shallow Tube well – 1, Filter point-2, Others - 3**
**Deep Tubewell : Deep Tube Well - 1, Others - 2**

**8. Details of the scheme**

( a ) Depth of the well (in metres) [ ][ ][ ]

( b ) Diameter **(unit in meters for dug well and mm for tube well )** [ ][ ]

( c ) Depth of Bore (in meters) **(in case of Dug-cum-borewell)** [ ][ ][ ]

( d) Distance of nearest well (in metres) [ ][ ][ ]

**9. (a) Cost of construction of the scheme** **(Rs.)** [ ][ ][ ][ ][ ][ ]

**(b) Cost of machinery** **(Rs.)** [ ][ ][ ][ ][ ][ ]

**(c) Cost of maintenance during (2006-07)** **(Rs.)** [ ][ ][ ][ ][ ][ ]

**10. Major source of finance** **Code :** [ ]

**Bank loan - 1, Government fund - 2, Own savings - 3, Money lender - 4, Others - 5**

**11. Current Status of the Scheme** **Code:** ☐

In use - 1, Temporarily Not in Use - 2, Permanently Not in use - 3

**12 . Reason code for Temporarily "not in use" Scheme (code - 2 in item 11)** **Code:** ☐

Non availability of adequate power – 1 , Mechanical break down – 2 ,
Less discharge in the well - 3, Any other reason - 4

**13 . Reason code for Permanently "not in use" Scheme (code - 3 in item 11)** **Code:** ☐

Due to salinity - 1, Dried up - 2 , Destroyed beyond repair - 3
Due to sea water intrusion - 4, Due to industrial effluents - 5,
Due to other reasons - 6

**14. Water distribution / application method used** **Code :** ☐

Open Water Channel (lined / pucca) – 1, Open Water Channel(unlined / kucha) - 2
Under ground channel – 3, Surface pipe - 4, Drip - 5 , Sprinkler - 6 , Other - 7

**15. Lifting device** **Code :** ☐

Submersible pump - 1 , Centrifugal Pump - 2, Turbine - 3, Manual/animal - 4, Other - 5

**16. Source of energy** **Code :** ☐

Electric - 1, Diesel - 2, Wind Mills - 3, Solar - 4, Manual/animal - 5, Others - 6

**17. Horse Power of Lifting device** ☐☐ **HP**

**(ignore if lifting device is manual/animal driven )**

**18. Number of days operating pump**

During Kharif season ☐☐☐ **Days**

During Rabi season ☐☐☐ **Days**

**19. Average hours of pumping per day**

During Kharif season ☐☐-☐☐ **Hrs**

During Rabi season ☐☐-☐☐ **Hrs**

**20. Culturable Command Area** ☐☐☐☐.☐☐ **Ha.**

**21. Whether the scheme is located in the command of Major & Medium Schemes like Canals etc.** **Code :** ☐

| | | |
|---|---|---|
| **No** | **--- 1** | **( Keep item 34 to item 38 blank)** |
| **Yes** | **--- 2** | **( Keep item 27 to item 31 blank)** |
| **Yes( for augumentation only)** | **--- 3** | **( Keep item 22 to item 38 blank)** |

## SEASON WISE IRRIGATION POTENTIAL CREATED (IPC)

22. **Kharif** [ ][ ][ ][ ].[ ][ ] Ha.
23. **Rabi** [ ][ ][ ][ ].[ ][ ] Ha.
24. **Perennial** [ ][ ][ ][ ].[ ][ ] Ha.
25. **Other** [ ][ ][ ][ ].[ ][ ] Ha.
26 **Total** [ ][ ][ ][ ].[ ][ ] Ha.

## SEASON WISE ACTUAL AREA IRRIGATED DURING 2006-07 (IPU)

27. **Kharif** [ ][ ][ ][ ].[ ][ ] Ha.
28. **Rabi** [ ][ ][ ][ ].[ ][ ] Ha.
29. **Perennial** [ ][ ][ ][ ].[ ][ ] Ha.
30. **Other** [ ][ ][ ][ ].[ ][ ] Ha.
31. **Total** [ ][ ][ ][ ].[ ][ ] Ha.

32. **Whether the scheme is functioning well since its commissioning ? ( Y / N )** Code : [ ]

33. **If No, Maximum Potential utilised from the Scheme in any year** [ ][ ][ ][ ].[ ][ ] Ha.

## SEASON WISE AREA IRRIGATED BY THE MINOR IRRIGATION SCHEMES (DURING 2006-07) AS SUPPLEMENTARY SOURCE IN THE COMMAND OF MAJOR & MEDIUM SCHEMES

34. **Kharif** [ ][ ][ ][ ].[ ][ ] Ha.
35. **Rabi** [ ][ ][ ][ ].[ ][ ] Ha.
36. **Perennial** [ ][ ][ ][ ].[ ][ ] Ha.
37. **Other** [ ][ ][ ][ ].[ ][ ] Ha.
38. **Total** [ ][ ][ ][ ].[ ][ ] Ha.

39. **Reasons for under utilisation of schemes** Code : [ ]

Non availability of adequate power – 1 , Mechanical break down – 2 ,
Less discharge in the well - 3 , Any other reason - 4 , Not applicable - 5

Checked by:
Name
Designation

Signature of Enumerator:
Name
Designation :

**FOURTH CENSUS OF MINOR IRRIGATION SCHEMES**
**REFERENCE YEAR 2006-07**

**SCHEDULE 2 : SURFACE WATER SCHEMES**

**I. IDENTIFICATION: (USE 2001 POPULATION CENSUS CODE ONLY)**

**State:** _____ **Code :** [ ][ ] **District:** _____ **Code:** [ ][ ]

**Block:** _____ **Code:** [ ][ ][ ] **Village:** _____ **Code:** [ ][ ][ ]

**Date of Enumeration:** [ ][ ] - [ ][ ] - [ ][ ]

**II. SPECIFIC INFORMATION:**

**1. Serial Number of scheme :** [ ][ ][ ]

**2. Type of Scheme** **Code:** [ ]

Surface Flow Scheme – 1, Surface Lift Scheme - 2

**3. Owner of the Scheme** ( Name in case of individual farmer)

Name _____ **Code:** [ ]

Govt. owned - 1, Co-operative owned - 2, Panchayat owned - 3, Owned by Group of farmers - 4,
Owned by individual farmers - 5, Others – 6

**4. Khasra number /Plot No./Survey No. in which the scheme is located** _____

**5(a). Total Holding of owner** (in case of individual owner only) [ ][ ][ ].[ ][ ] **Ha.**

**(b) Social Status of Owner** ( in case of individual owner only) **Code:** [ ]

Scheduled caste -1 , Scheduled tribe – 2 , OBC- 3, Others 4

**6. Year of Commissioning of the Scheme** **Code :** [ ]

On or before 2000-2001 - 1, during 2001-02 - 2, during 2002-03 - 3, during 2003-04 - 4
during 2004-05 - 5 , during 2005-06 - 6 , during 2006-07 - 7

**7. Nature of scheme** **Code :** [ ]

**Surface Flow Scheme : Tanks -1, Other Storages -2, Permanent diversion - 3, Temporary diversion - 4**
**Water conservation-cum-ground water recharge schemes /percolation tanks/check dams etc - 5**
**Spring Channel - 6, Other - 7**

**Surface lift Scheme : On River - 1, On Stream - 2, On drain/canal - 3 , On Tanks/Ponds - 4, Others - 5**

**8. (a) Cost of construction of the scheme** **(Rs.)** [ ][ ][ ][ ][ ][ ]

**(b) Cost of machinery** **(Rs.)** [ ][ ][ ][ ][ ][ ]

**(c) Cost of maintenance during (2006-07)** **(Rs.)** [ ][ ][ ][ ][ ][ ]

**9. Major source of finance** **Code :** [ ]

**Bank loan - 1, Government fund - 2, Own savings - 3, Money lender - 4, Others - 5**

**10. Current Status of the Scheme** **Code:** ☐

In use - 1, Temporarily Not in Use - 2, Permanently Not in use - 3

**11 . Reason code for Temporarily "not in use" Scheme (code - 2 in item 10)** **Code:** ☐

Non availability of adequate power – 1 , Mechanical break down – 2 ,
Less discharge of water - 3 , Storage not filled up fully - 4
Siltation of canal/storage - 5, Breakdown of channels - 6, Any other reason - 7

**12 . Reason code for Permanently "not in use" Scheme (code - 3 in item 10)** **Code:** ☐

Due to salinity - 1, Dried up - 2 , Destroyed beyond repair - 3,
Due to sinking - 4, Due to other reasons - 5

**13. Water distribution / application method** **Code :** ☐

Open Water Channel (lined / pucca) – 1, Open Water Channel(unlined / kucha) - 2
Under ground channel – 3, Surface pipe - 4, Drip - 5 , Sprinkler - 6 , Others -7

**14. Lifting device ( in case of Surface lift scheme )** **Code :** ☐

Submersible pump - 1 , Centrifugal Pump - 2, Turbine - 3, Manual/animal - 4, Other - 5

**15. Source of energy** **Code :** ☐

Electric - 1, Diesel - 2, Wind Mills - 3, Solar - 4, Manual/animal - 5, Others - 6

**16. Horse Power of Lifting device ( in case of Surface lift scheme )** ☐☐ **HP**

**17. Number of days operating pump**

During Kharif season ☐☐☐ **Days**

During Rabi season ☐☐☐ **Days**

**18. Average hours of pumping per day**

During Kharif season ☐☐☐-☐☐ **Hrs**

During Rabi season ☐☐☐-☐☐ **Hrs**

**19. Culturable Command Area** ☐☐☐☐.☐☐ **Ha.**

**20. Whether the scheme is located in the command of Major & Medium Schemes like Canals etc.** **Code :** ☐

**No** --- 1 **( Keep item 33 to item 37 blank)**
**Yes** --- 2 **( Keep item 26 to item 30 blank)**
**Yes(for augumentation only)** --- 3 **( Keep item 21 to item 37 blank)**

## SEASON WISE IRRIGATION POTENTIAL CREATED (IPC)

21. **Kharif** [ ][ ][ ][ ].[ ][ ] Ha.
22. **Rabi** [ ][ ][ ][ ].[ ][ ] Ha.
23. **Perennial** [ ][ ][ ][ ].[ ][ ] Ha.
24. **Other** [ ][ ][ ][ ].[ ][ ] Ha.
25. **Total** [ ][ ][ ][ ].[ ][ ] Ha.

## SEASON WISE ACTUAL AREA IRRIGATED DURING 2006-07 ( IPU )

26. **Kharif** [ ][ ][ ][ ].[ ][ ] Ha.
27. **Rabi** [ ][ ][ ][ ].[ ][ ] Ha.
28. **Perennial** [ ][ ][ ][ ].[ ][ ] Ha.
29. **Other** [ ][ ][ ][ ].[ ][ ] Ha.
30. **Total** [ ][ ][ ][ ].[ ][ ] Ha.

31. **Whether the scheme is functioning well since its commissioning ? ( Y / N )** Code : [ ]

32. **If No, Maximum Potential utilised from the Scheme in any year** [ ][ ][ ][ ].[ ][ ] Ha.

## SEASON WISE AREA IRRIGATED BY THE MINOR IRRIGATION SCHEMES (DURING 2006-07) AS SUPPLEMENTARY SOURCE IN THE COMMAND OF MAJOR & MEDIUM SCHEMES

33. **Kharif** [ ][ ][ ][ ].[ ][ ] Ha.
34. **Rabi** [ ][ ][ ][ ].[ ][ ] Ha.
35. **Perennial** [ ][ ][ ][ ].[ ][ ] Ha.
36. **Other** [ ][ ][ ][ ].[ ][ ] Ha.
37. **Total** [ ][ ][ ][ ].[ ][ ] Ha.

38. **Reasons for under utilisation of schemes** Code: [ ]

Non availability of adequate power – 1 , Mechanical break down – 2 ,
Less discharge of water - 3 , Storage not filled up fully - 4, Siltation of canal/storage- 5
Breakdown of channels - 6, Any other reason - 7, Not applicable - 8

39. **Some features of Tank / Other storages**

( a ) Designed Storage **(in cubic metres)** [ ][ ][ ][ ]

( b ) Filled up Storage **( during 2006-07)** Code : [ ]

Full - 1 , 3/4 filled up - 2 , 1/4 filled up - 3 , Never filled up - 4

( c ) Status of filling up of storage Space Code : [ ]

Filled up every year - 1 , Usually filled up - 2 , Rarely filled up - 3 , Never filled up - 4

( d ) Number of Villages covered by the scheme [ ][ ]

40. **Whether the Scheme has benifited Ground Water in the area** Code : [ ]

Yes - 1 , No - 2 , Can't say - 3

Checked by:
Name
Designation

Signature of Enumerator:
Name
Designation

FOURTH CENSUS OF MINOR IRRIGATION SCHEMES
REFERENCE YEAR 2006-07

## VILLAGE SCHEDULE

**I. IDENTIFICATION: (USE 2001 POPULATION CENSUS CODE ONLY)**

State: _____ Code : [ ][ ] District: _____ Code: [ ][ ]

Block: _____ Code: [ ][ ][ ] Village: _____ Code : [ ][ ][ ]

Date of Enumeration: [ ][ ] - [ ][ ] - [ ][ ]

**II. SPECIFIC INFORMATION:**

1. Is Village Tribal/ Non-Tribal ? Code : [ ]
   Tribal – 1 , Non-Tribal -2

2. Name of Major/ Medium Scheme _____

(Note : The information in item 3 to 11 of this schedule shall be based on village records)

3. Geographical Area [ ][ ][ ][ ].[ ][ ] Ha.
4. Cultivable Area [ ][ ][ ][ ].[ ][ ] Ha.
5. Net Area Sown [ ][ ][ ][ ].[ ][ ] Ha.
6. Net Area Irrigated through Canal [ ][ ][ ][ ].[ ][ ] Ha.
7. Net Area Irrigated through Tanks [ ][ ][ ][ ].[ ][ ] Ha.
8. Net Area Irrigated through Tube wells [ ][ ][ ][ ].[ ][ ] Ha.
9. Net Area Irrigated through other wells [ ][ ][ ][ ].[ ][ ] Ha.
10. Net Area Irrigated through others sources [ ][ ][ ][ ].[ ][ ] Ha.
11. Total Net Area Irrigated (column 6+7+8+9+10) [ ][ ][ ][ ].[ ][ ] Ha.
12. Average Ground Water level (in Meters) [ ][ ][ ][ ].[ ][ ]
13. Number of waterbody (a) for irrigation [ ][ ][ ][ ]
    (b) for non irrigation [ ][ ][ ][ ]
14. Whether Water Users Association (WUA) exists in the village Code: [ ]
    Yes -1, No -2
    If yes , Number of Water Users Associations : [ ][ ][ ]

---

Schedule wise Summary of M I schemes in the village

GROUND WATER SCHEDULES [ ][ ][ ][ ]

SURFACE WATER SCHEDULES [ ][ ][ ][ ]

TOTAL SCHEDULES [ ][ ][ ][ ]

Checked by:
Name
Designation of Supervisory officer:

Signature of Enumerator:
Name
Designation of Enumerator :

## SEASON WISE IRRIGATION POTENTIAL CREATED (IPC)

21. **Kharif** [ ][ ][ ][ ].[ ][ ] Ha.
22. **Rabi** [ ][ ][ ][ ].[ ][ ] Ha.
23. **Perennial** [ ][ ][ ][ ].[ ][ ] Ha.
24. **Other** [ ][ ][ ][ ].[ ][ ] Ha.
25. **Total** [ ][ ][ ][ ].[ ][ ] Ha.

## SEASON WISE ACTUAL AREA IRRIGATED DURING 2006-07 ( IPU )

26. **Kharif** [ ][ ][ ][ ].[ ][ ] Ha.
27. **Rabi** [ ][ ][ ][ ].[ ][ ] Ha.
28. **Perennial** [ ][ ][ ][ ].[ ][ ] Ha.
29. **Other** [ ][ ][ ][ ].[ ][ ] Ha.
30. **Total** [ ][ ][ ][ ].[ ][ ] Ha.

31. **Whether the scheme is functioning well since its commissioning ? ( Y / N )** Code : [ ]

32. **If No, Maximum Potential utilised from the Scheme in any year** [ ][ ][ ][ ].[ ][ ] Ha.

## SEASON WISE AREA IRRIGATED BY THE MINOR IRRIGATION SCHEMES (DURING 2006-07) AS SUPPLEMENTARY SOURCE IN THE COMMAND OF MAJOR & MEDIUM SCHEMES

33. **Kharif** [ ][ ][ ][ ].[ ][ ] Ha.
34. **Rabi** [ ][ ][ ][ ].[ ][ ] Ha.
35. **Perennial** [ ][ ][ ][ ].[ ][ ] Ha.
36. **Other** [ ][ ][ ][ ].[ ][ ] Ha.
37. **Total** [ ][ ][ ][ ].[ ][ ] Ha.

38. **Reasons for under utilisation of schemes** Code: [ ]

Non availability of adequate power – 1 , Mechanical break down – 2 ,
Less discharge of water - 3 , Storage not filled up fully - 4, Siltation of canal/storage- 5
Breakdown of channels - 6, Any other reason - 7, Not applicable - 8

39. **Some features of Tank / Other storages**

( a ) Designed Storage **(in cubic metres)** [ ][ ][ ][ ]

( b ) Filled up Storage **( during 2006-07)** Code : [ ]

Full - 1 , 3/4 filled up - 2 , 1/4 filled up - 3 , Never filled up - 4

( c ) Status of filling up of storage Space Code : [ ]

Filled up every year - 1 , Usually filled up - 2 , Rarely filled up - 3 , Never filled up - 4

( d ) Number of Villages covered by the scheme [ ][ ]

40. **Whether the Scheme has benifited Ground Water in the area** Code : [ ]

Yes - 1 , No - 2 , Can't say - 3

Checked by:
Name
Designation

Signature of Enumerator:
Name
Designation